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Portugal Hotel Business Report

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1. Introduction

1.1 Business Context & Data Source

This report provides a comprehensive ML-based Business Analytics overview for the hotel industry in Portugal, drawing upon research conducted by Nuno Antonio, Ana de Almeida, and Luis Nunes. The source data comprises two datasets that detail hotel booking information for a resort and a city hotel in Portugal. Each dataset includes over 30 variables describing thousands of bookings, covering both confirmed and canceled reservations between July 2015 and August 2017. Notably, all personally identifying information has been removed.

The goal of this report is to leverage conventional machine learning (ML) models to extract patterns and insights from the dataset, ultimately generating business suggestions and strategies for stakeholders such as hotel managers and owners.

The primary research areas include:

- **Customer Booking cancelation pattern:** This include using clustering model such as hierarchical clustering, classifying model such as decision trees, Naïve Bayesian, SVM(SVC) to potentially detect customer cancelation.
- **Room price optimization:** Applying several regression models, including OLS, GLS, GLSAR, and OLS-HAC, to potentially optimize the average daily rate (ADR). The aim is to gain insights from the independent variables that influence hotel room prices and subsequently provide business strategies.
- **Customer purchasing pattern:** Determining common booking combinations among customers. This research problem will be addressed using association rule mining to identify lift and other rule attributes, generating further business insights.

- **Customer segmentation:** Primarily, I will use the hierarchical clustering method to determine the number of customer segments within Portugal's hotels. This information will inform marketing strategies, helping hotels to target their customers more effectively.

1.2 Data Cleaning and Processing

The initial size of the dataset contains 119390 records.

Here are all the variables for the hotel booking dataset for referencing:

- **ADR (Average Daily Rate):** This is a numeric variable, calculated by dividing the sum of all lodging transactions by the total number of staying nights.
- **Adults:** An integer representing the number of adults in the booking.
- **Agent:** A categorical variable indicating the ID of the travel agency that made the booking.
- **ArrivalDateDayOfMonth:** An integer representing the day of the month of the arrival date.
- **ArrivalDateMonth:** A categorical variable representing the month of the arrival date, with 12 categories from "January" to "December".
- **ArrivalDateWeekNumber:** An integer representing the week number of the arrival date.
- **ArrivalDateYear:** An integer representing the year of the arrival date.
- **AssignedRoomType:** A categorical variable showing the code for the type of room assigned to the booking. Note that the assigned room type can differ from the reserved type.

- **Babies:** An integer representing the number of babies in the booking.
- **BookingChanges:** An integer representing the number of changes/amendments made to the booking.
- **Children:** An integer representing the number of children in the booking.
- **Company:** A categorical variable indicating the ID of the company that made the booking.
- **Country:** A categorical variable representing the country of origin, using ISO 3155–3:2013 format.
- **CustomerType:** A categorical variable indicating the type of booking, with four categories: Contract, Group, Transient, and Transient-party.
- **DaysInWaitingList:** An integer representing the number of days the booking was in the waiting list before confirmation.
- **DepositType:** A categorical variable indicating if the customer made a deposit, with categories: No Deposit, Non Refund, and Refund.
- **DistributionChannel:** A categorical variable indicating the booking distribution channel, such as Travel Agents (TA), Tour Operators (TO), and Direct.
- **IsCanceled:** A categorical variable indicating if the booking was canceled (1) or not (0).
- **IsRepeatedGuest:** A categorical variable indicating if the booking name was from a repeated guest (1) or not (0).

- **LeadTime:** An integer representing the number of days between the booking date and the arrival date.
- **MarketSegment:** A categorical variable indicating the market segment designation, with terms like "TA" for "Travel Agents" and "TO" for "Tour Operators".
- **Meal:** A categorical variable indicating the type of meal booked, with categories such as Undefined/SC (no meal package), BB (Bed & Breakfast), HB (Half board), and FB (Full board).
- **PreviousBookingsNotCanceled:** An integer representing the number of previous bookings not canceled by the customer.
- **PreviousCancellations:** An integer representing the number of previous bookings canceled by the customer.
- **RequiredCardParkingSpaces:** An integer representing the number of car parking spaces required by the customer.
- **ReservationStatus:** A categorical variable indicating the last status of the reservation, with categories such as Canceled, Check-Out, and No-Show.
- **ReservationStatusDate:** A date variable indicating the date at which the last status was set.
- **ReservedRoomType:** A categorical variable indicating the code of the room type reserved.
- **StaysInWeekendNights:** An integer representing the number of weekend nights the guest stayed or booked to stay at the hotel.

- **StaysInWeekNights:** An integer representing the number of week nights the guest stayed or booked to stay at the hotel.
- **TotalOfSpecialRequests:** An integer representing the number of special requests made by the customer.

There are two essential cleaning steps for this hotel booking dataset. First, I removed 26.8% (32,001 records) of the duplicate rows. I suspect, based on the original articles, that this issue stemmed from SQL database duplication during the hotel booking process, where customers frequently changed their booking details, creating numerous duplicate entries. This suggests a potentially flawed system database design within Portuguese hotel booking systems. Second, I removed outliers using the Median Absolute Deviation (MAD) method, which is particularly suitable for this dataset due to the high skewness of its variables. Addressing outliers and duplicate values is crucial at this stage as it enhances the accuracy and reliability of the model's results. After these two steps, the dataset was reduced to a total of 71,368 records.

The remaining minor issues involved missing values in the 'country' and 'agent' columns, both of which are categorical. To address this, I labeled all missing values as 'unknown,' effectively creating a new category. This approach is justified because the number of missing values is relatively small compared to the overall dataset size. Labeling them as 'unknown' preserves information for the model and is particularly beneficial for subsequent clustering implementations.

Given the abundance of categorical variables, I will create dummy variables for all relevant columns to facilitate model analysis. The number of columns will vary throughout the report due

to limitations in certain models' capacity to handle a large number of variables. The most frequently used dataset information is as follows:

Records: 71,368, Columns: 93

Specific dataset details for each model will be mentioned in their respective analyses.

2. Exploratory Data Analysis (EDA)

2.1 Descriptive Statistics

A key characteristic of this dataset is its skewness.

I observed that several variables, particularly ['number_of_children'], exhibited a high degree of skewness. For example, the 'number_of_children' variable frequently had a value of 0, indicating a rightward skewness in the data (+/- skewness value for the data). This distribution challenges the assumptions of Ordinary Least Squares (OLS) regression (which I will implement in a future report), as OLS relies on the assumption that the residuals are normally distributed and have constant variance.

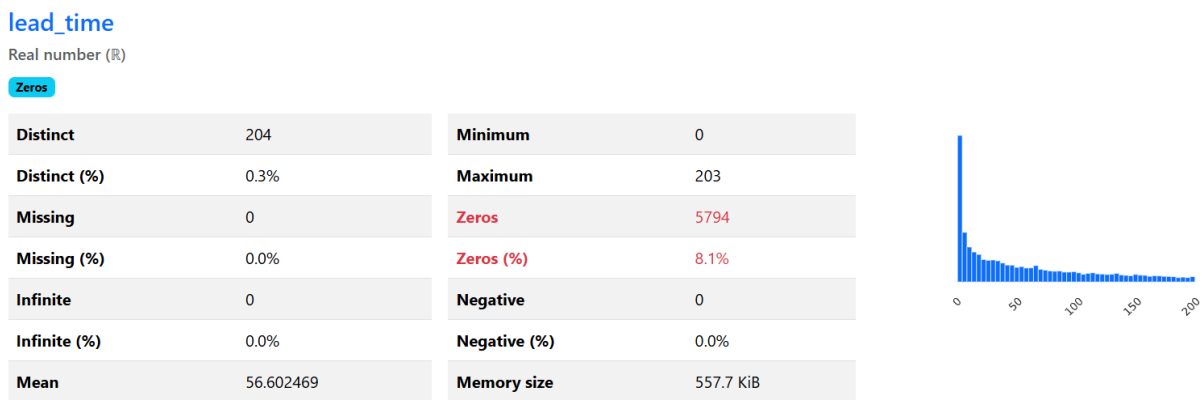


Figure 1: The example of lead time information

Another example of right skewness can be seen in Figure 1. In this case, the mean lead time is 56, yet the total range (after outlier removal) spans from approximately 0 to 203, further demonstrating the rightward skew of the dataset.

2.2 Preliminary Insights

Here are some important variables that I believe are worth mentioning in this EDA:

Is_canceled

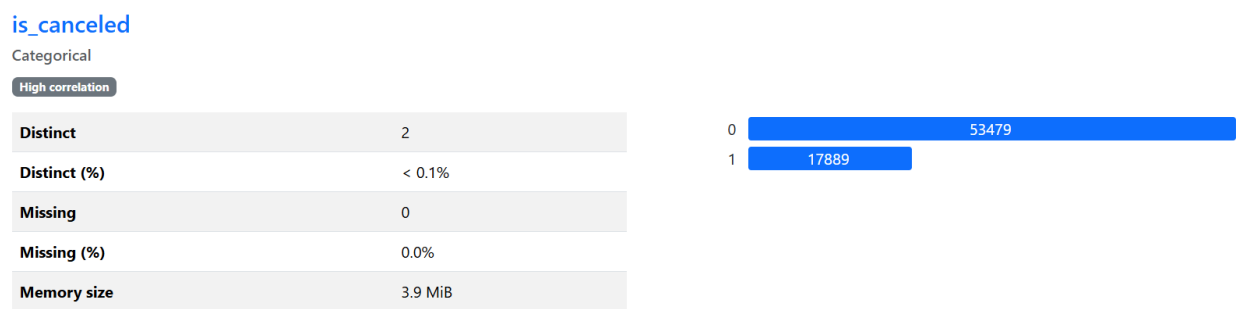


Figure 2.1: the canceled distribution for the dataset

This is an important column that will be used extensively for all the models because my goal is to determine which customers will cancel their bookings; therefore, this column will typically be my target variable. It is worth noting that the proportion of cancellations (1) and non-cancellations (0) is imbalanced. To avoid calculation bias in further model analysis, I need to take this into consideration.

Adr

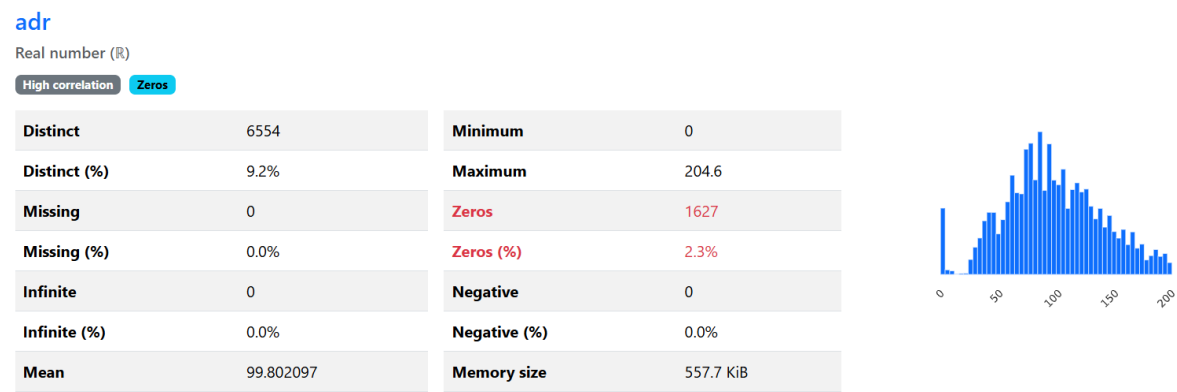


Figure 2.2 descriptive analysis for adr (average room price)

The ADR (Average Daily Rate) represents the price of the hotel room, which will be the dependent variable in my regression model analysis. It appears to be normally distributed and could provide some valuable insights into the Portuguese hotel market.

Country

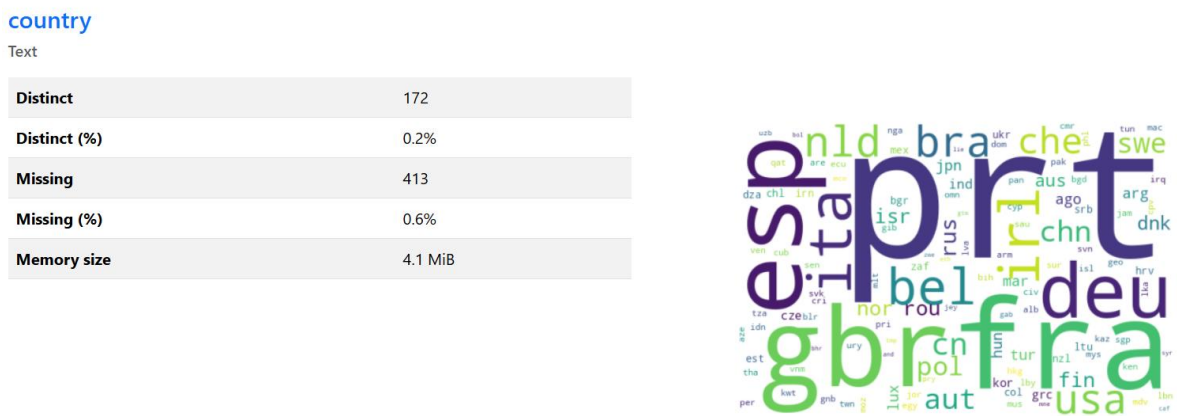


Figure 2.3: country column descriptive analysis

The "country" column will be implemented in later model analysis, primarily because it is a significant feature influencing customer purchasing decisions and cancellation rates.

3. Regression Analysis

OLS Model

In my regression analysis, I used an ordinary least squares (OLS) model to analyze the impact of selected variables on the average daily rate (ADR).

The results are as follows:

Model Summary:			
OLS Regression Results			
=====			
Dep. Variable:	adr	R-squared:	0.608
Model:	OLS	Adj. R-squared:	0.608
Method:	Least Squares	F-statistic:	1783.
Date:	Thu, 27 Feb 2025	Prob (F-statistic):	0.00
Time:	16:50:52	Log-Likelihood:	-3.3603e+05
No. Observations:	71368	AIC:	6.722e+05
Df Residuals:	71305	BIC:	6.728e+05
Df Model:	62		
Covariance Type:	nonrobust		

Figure 3.1: Model Summary of OLS

Figure 3.1 shows an R-squared value of approximately 0.608, indicating that the model explains a moderate amount of the variance in the outcome. This is acceptable, given the inherent variability in hotel room prices due to factors like time of year and holidays. The F-statistic and its associated p-value also appear satisfactory, suggesting no immediate cause for concern.

	coef	std err	t	P> t	[0.025	0.975]
const	67.6135	1.540	43.906	0.000	64.595	70.632
is_canceled	5.8411	0.256	22.804	0.000	5.339	6.343
lead_time	-0.1219	0.002	-55.577	0.000	-0.126	-0.118
arrival_date_day_of_month	0.1246	0.011	10.921	0.000	0.102	0.147
stays_in_weekend_nights	0.3587	0.145	2.479	0.013	0.075	0.642
stays_in_week_nights	1.4223	0.086	16.563	0.000	1.254	1.591
adults	13.1645	0.228	57.765	0.000	12.718	13.611
children	12.9172	0.298	43.409	0.000	12.334	13.500
babies	1.7363	0.903	1.922	0.055	-0.034	3.507
is_repeated_guest	-12.8963	0.583	-22.114	0.000	-14.039	-11.753
previous_cancellations	0.9620	0.348	2.766	0.006	0.280	1.644
previous_bookings_not_canceled	-0.3583	0.067	-5.321	0.000	-0.490	-0.226
booking_changes	0.8506	0.157	5.416	0.000	0.543	1.158
days_in_waiting_list	0.2057	0.020	10.358	0.000	0.167	0.245
required_car_parking_spaces	6.7711	0.374	18.084	0.000	6.037	7.505
total_of_special_requests	1.2067	0.136	8.880	0.000	0.940	1.473
is_city_hotel	28.0739	0.250	112.213	0.000	27.584	28.564
meal_FB	41.8060	1.851	22.582	0.000	38.177	45.435
meal_HB	24.4064	0.396	61.574	0.000	23.629	25.183
meal_SC	-13.5763	0.343	-39.573	0.000	-14.249	-12.904
meal_undefined	35.1581	1.473	23.868	0.000	32.271	38.045
market_segment_Aviation	-10.6269	1.911	-5.561	0.000	-14.372	-6.881
market_segment_Complementary	-101.6043	1.059	-95.907	0.000	-103.681	-99.528
market_segment_Corporate	-16.6901	0.531	-31.452	0.000	-17.730	-15.650
market_segment_Direct	-2.5214	0.323	-7.801	0.000	-3.155	-1.888

market_segment_Groups	-26.1538	0.615	-42.540	0.000	-27.359	-24.949
market_segment_Offline TA/TO	-24.6879	0.328	-75.221	0.000	-25.331	-24.045
market_segment_undefined	-120.9602	19.001	-6.366	0.000	-158.203	-83.718
assigned_room_type_B	-4.1246	0.721	-5.719	0.000	-5.538	-2.711
assigned_room_type_C	6.8591	0.731	9.384	0.000	5.426	8.292
assigned_room_type_D	9.0905	0.252	36.063	0.000	8.596	9.585
assigned_room_type_E	18.4418	0.415	44.484	0.000	17.629	19.254
assigned_room_type_F	26.5906	0.598	44.435	0.000	25.418	27.764
assigned_room_type_G	26.7560	0.789	33.908	0.000	25.209	28.303
assigned_room_type_H	24.5941	1.370	17.949	0.000	21.908	27.280
assigned_room_type_I	-37.1653	1.634	-22.750	0.000	-40.367	-33.963
assigned_room_type_K	-32.4830	1.776	-18.292	0.000	-35.964	-29.002
assigned_room_type_L	-89.8216	26.848	-3.346	0.001	-142.444	-37.199
assigned_room_type_P	-11.5088	12.027	-0.957	0.339	-35.081	12.064
deposit_type_No Deposit	-11.7777	1.227	-9.598	0.000	-14.183	-9.373
deposit_type_Refundable	10.3511	3.154	3.282	0.001	4.169	16.533
customer_type_Contract	-1.7083	0.706	-2.421	0.015	-3.091	-0.325
customer_type_Group	-1.5733	1.299	-1.211	0.226	-4.119	0.973
customer_type_Transient	-1.7158	0.372	-4.610	0.000	-2.445	-0.986
year_2015	-25.8657	0.402	-64.377	0.000	-26.653	-25.078
year_2016	-12.8938	0.247	-52.281	0.000	-13.377	-12.410
month_1	-20.2793	0.604	-33.590	0.000	-21.463	-19.096
month_2	-17.0552	0.565	-30.161	0.000	-18.164	-15.947
month_3	-8.5685	0.543	-15.793	0.000	-9.632	-7.505
month_4	10.0963	0.542	18.645	0.000	9.035	11.158
month_5	22.2252	0.547	40.628	0.000	21.153	23.297
month_6	29.3915	0.560	52.518	0.000	28.295	30.488
month_7	37.7493	0.529	71.343	0.000	36.712	38.786

```

month_8          41.4555      0.522      79.353      0.000      40.432      42.479
month_9          36.9121      0.554      66.644      0.000      35.827      37.998
month_10         15.4256      0.536      28.770      0.000      14.375      16.477
month_11         -2.2427      0.567      -3.954      0.000      -3.354      -1.131
week_0           -1.3483      0.391      -3.445      0.001      -2.115      -0.581
week_1           -1.9225      0.434      -4.433      0.000      -2.772      -1.073
week_2           -2.0777      0.408      -5.089      0.000      -2.878      -1.278
week_3           -1.8681      0.385      -4.854      0.000      -2.622      -1.114
week_4           -1.6172      0.374      -4.320      0.000      -2.351      -0.883
week_5           -1.8663      0.391      -4.776      0.000      -2.632      -1.100
=====
Omnibus:          4306.089      Durbin-Watson:      1.162
Prob(Omnibus):    0.000      Jarque-Bera (JB):   19367.072
Skew:             0.046      Prob(JB):           0.00
Kurtosis:         5.550      Cond. No.           2.16e+04
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 2.16e+04. This might indicate that there are
strong multicollinearity or other numerical problems.

Durbin-Watson statistic: 1.1616201571748364

```

Figure 3.2: statistics of OLS

Figure 3.2 presents diagnostic statistics focusing on the p-values, Durbin-Watson test, and Jarque-Bera (JB) test to assess the model's residuals. Several dummy variables, such as 'assign_room_type_P' and 'customer_type_group', exhibit p-values exceeding 0.05, indicating they are not statistically significant. However, their interpretation requires consideration of their role as dummy variables. 'assign_room_type_P' was included by design to compare room type P against a reference category (room type A) to prevent multicollinearity (avoiding infinite VIF). Its non-significant p-value suggests insufficient evidence to conclude a price difference between room types P and A, controlling for other variables. Similar reasoning applies to 'customer_type_group'. Therefore, these variables will be retained, but their interpretation will require careful attention.

The non-dummy variable 'babies' is also statistically insignificant. This discrete variable, skewed due to a high proportion of zero values (likely because few hotel bookings in Portugal involve

babies), may be addressed through a transformation or removal. A log transformation is unsuitable for zero values, even with adjustments. Removal might be preferable, considering the prevalence of zero values, suggesting limited analytical utility.

The Durbin-Watson statistic of 1.162 hints at potential positive autocorrelation among the residuals, which is not unexpected given the use of time-series data. Finally, the JB and Omnibus tests both yield p-values of 0, suggesting a significant departure from normality in the residuals.

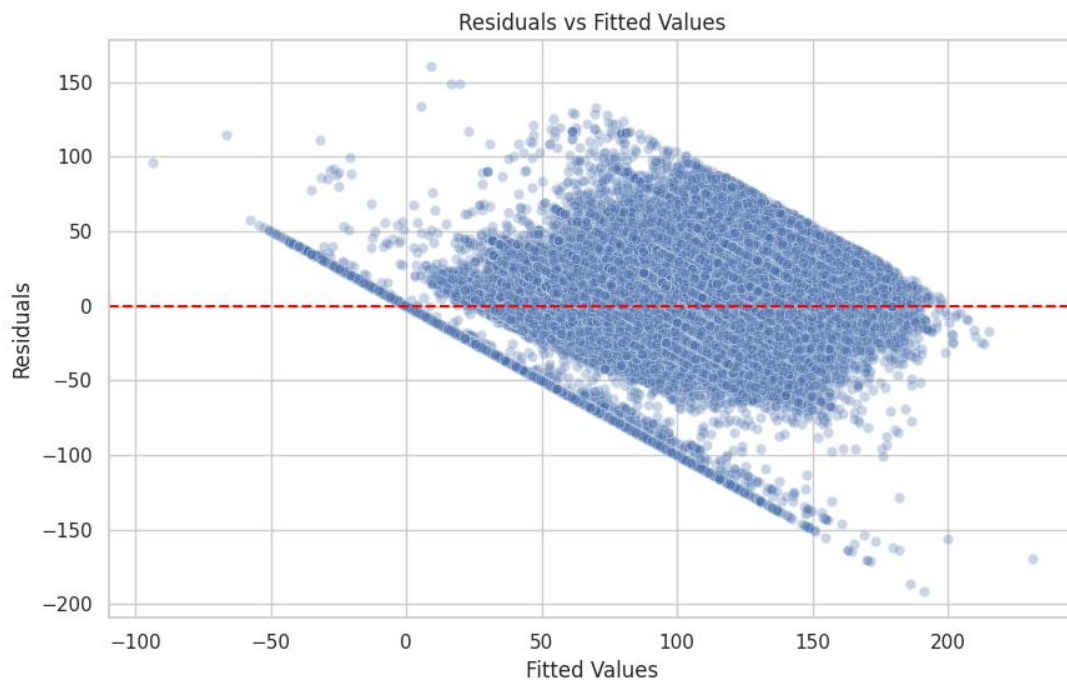


Figure 3.3: Graph of scatter plot that shows autocorrelation of the residuals

Figure 3.3 shows a plot of residuals against fitted values, reveals evidence of heteroskedasticity; the residuals' spread is not constant across the range of fitted values.

```
Breusch-Pagan test results:  
Lagrange multiplier statistic: 6942.968668596019  
p-value: 0.0  
f-value: 123.94210325857085  
f p-value: 0.0
```

Figure 3.4: Breusch-Pagan test results for heteroskedasticity

Figure 3.4's Breusch-Pagan (BP) test confirms the presence of heteroskedasticity, indicated by a low p-value. This is consistent with the observation of non-constant residual variance in Figure 3.3. The varying price ranges across different room types (e.g., standard versus luxury rooms) likely contribute to this heteroskedasticity, as price fluctuations might be more pronounced for higher-priced rooms. Furthermore, the indication of positive autocorrelation from the Durbin-Watson test suggests potential non-linearity in the relationship (e.g., seasonality), which could also be a source of heteroskedasticity.

```
Shapiro-Wilk test statistic: 0.9720780700784579, p-value: 3.652017443851052e-30
```

Figure 3.5: Shapiro-Wilk test statistic introduces unnormal residuals

Figure 3.5, showing the results of the Shapiro-Wilk test, indicates a departure from normality in the residuals. The low p-value from the Shapiro-Wilk test confirms this, visually supported by the deviations from the diagonal in the Q-Q plot (figure 3.6):

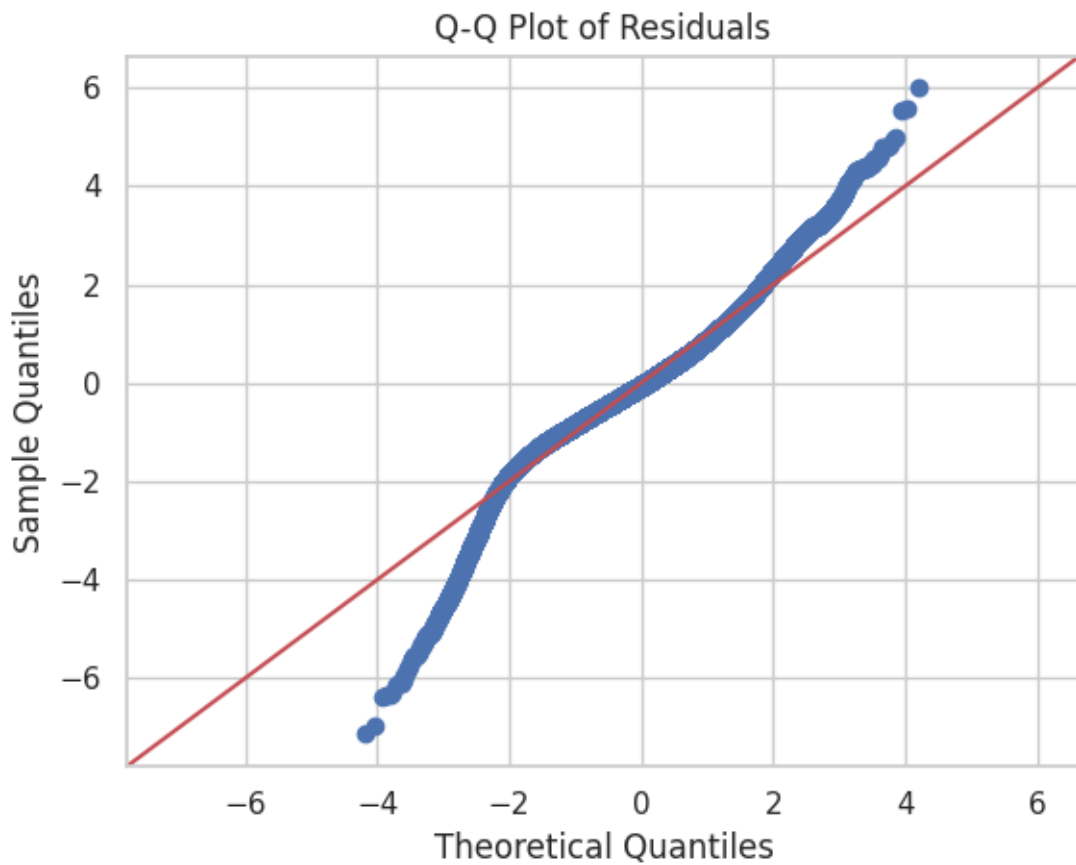


Figure 3.6: Q-Q plot for checking normality of residuals

The S-shaped pattern of residuals in Figure 3.6 suggests potential non-linearity in the model, possibly due to seasonal effects or other systematic patterns. This non-linearity is another contributing factor to the non-normality of the residuals.

The issue of multicollinearity, as addressed by the Variance Inflation Factor (VIF) test, has been resolved and will not be further discussed.

Key things need to be addressed

The key issues that need to be addressed here are quite obvious. Switching to another model that better fits the non-normal distribution would be preferable, especially when dealing with seasonality. Additionally, I need to find a model that can address autocorrelation.

Considering all these issues, this model still provides a decent understanding of the relationship with 'adr' based on p-value. However, the reliability or robustness of the outcome is very questionable.

OLS-HAC

Another model I implemented was the OLS-HAC model, which specifically addresses the issues of autocorrelation and heteroscedasticity in time series data. Because this model builds upon the original OLS, I only adjusted the maxlag for the OLS-HAC model to 37 for a more robust adjustment. The maxlag was calculated using the equation: $12 * (713.68)^{1/4} \approx 37$, where 713.68 represents the current data size divided by 100. This model should be more robust than the original OLS model, and we will be comparing them to identify the most robust model in further model comparison.

GLS

I implemented GLS for my model, but it didn't fix any problems; everything remained the same. Even the BP test result was unchanged, indicating that implementing GLS didn't resolve the heteroskedasticity. Therefore, implementing the GLS model was unsuccessful, and as a result, we can still stick to the original OLS model (Assuming OLS-HAC and GLSAR didn't exist).

GLSAR(next step)

Since implementing GLS didn't fix the issues, I implemented the GLSAR model, which might be a better solution for autocorrelation in time series data. The key to this model is reordering the records based on timeframe to allow the model to compute correctly. So, I reordered the dataset based on the 'reservation_status_date' column. I also set the `ar_order = 1`, which is the current default order.

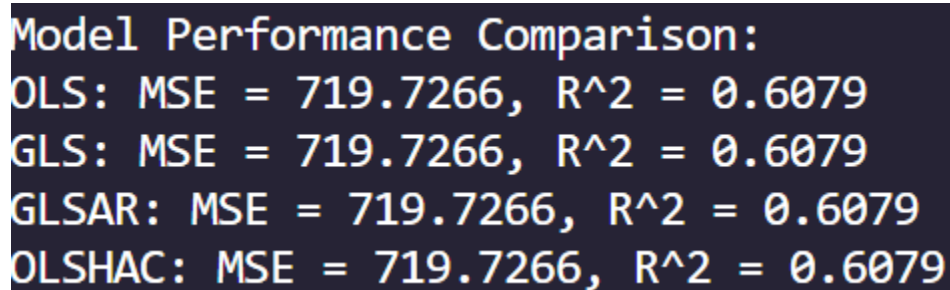
Here's the Durbin-Watson test result before and after:

GLS (before): 1.164

GLSAR (after): 1.638

After implementing the GLSAR model, the autocorrelation has significantly improved, which is very promising. This indicates that the GLSAR model does help with the autocorrelation issues. However, the autocorrelation is still not close to 2, suggesting further adjustment is needed. I attempted to change `ar_order=2` to see what would happen, but an internal codebase error occurred in statsmodels, so I had to move on to another model. So far, the GLSAR model is the best model in terms of addressing autocorrelation.

Model comparison



```
Model Performance Comparison:
OLS: MSE = 719.7266, R^2 = 0.6079
GLS: MSE = 719.7266, R^2 = 0.6079
GLSAR: MSE = 719.7266, R^2 = 0.6079
OLSHAC: MSE = 719.7266, R^2 = 0.6079
```

Figure 4: The side-by-side model comparison

In model comparison, all the MSE and R-squared values remained constant. In theory, GLSAR could be the most robust model for handling the current hotel dataset. However, due to a lack of codebase and calculation knowledge regarding GLSAR (specifically, a lack of experience in AR configuration), I would still consider the OLS-HAC model the most trustworthy and robust model for this dataset at this time. This is because it addresses the autocorrelation problem inherent in the time series data and also takes heteroskedasticity into consideration, making it more robust than OLS.

In conclusion, I will use the results of the OLS-HAC model for creating business inside for the report.

4. Classification / Prediction

This section presents classification and prediction results for customer cancellations, using Naïve Bayes, decision trees, and support vector machines.

4.1 Naïve Bayes

Methodology

Five rounds of testing will compare GaussianNB (for continuous variables), MultinomialNB (for discrete variables), and BernoulliNB (for categorical variables) to identify the optimal Naïve Bayes model for classifying hotel booking cancellations. Model selection will be based on AUC, accuracy, and F1-score. The training and testing data will be split 75%/25%, respectively. The testing process and rationale will be detailed in this section.

Process and Outcomes

Dumping every variable (first test)

I started by using GaussianNB to conduct my initial model test. The idea was to input all available data from the dataset as features into the model. Consequently, the X features contained both discrete and continuous variables, meaning the model's performance is based on a mix of variable types. I bring this up because GaussianNB assumes that the variables are normally distributed. Given that I have some count variables and binomial variables (0, 1), I wasn't sure what the model's behavior would be. Fortunately, the scikit-learn library's GaussianNB implementation can accept these variables, so we will see what happens.

Here is the outcome:

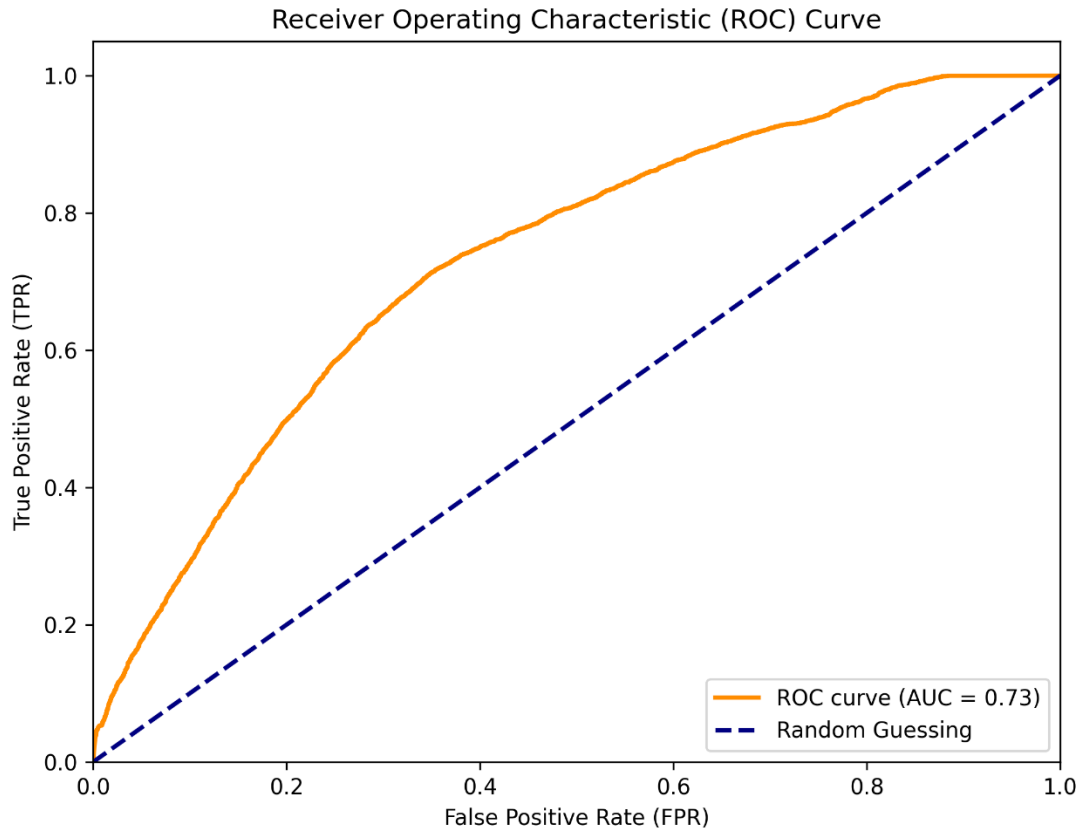


Figure 5.1: ROC-AUC of test 1 result

Figure 5.1 shows an AUC of 0.73, which is in a moderate range for the first model. This result is expected, given the nature of using a mixture of features in a GaussianNB model. I will use this AUC as a baseline to compare other tests later in this report and see how they perform, but for now, the performance is looking moderate.

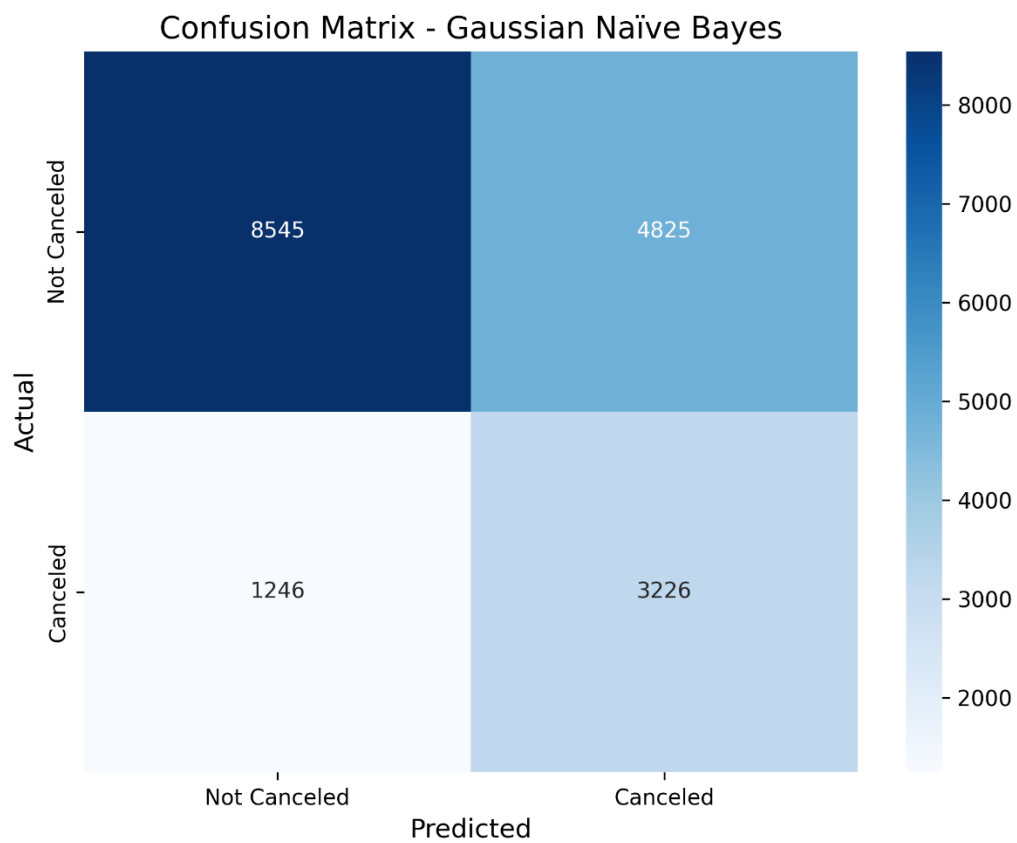


Figure 5.2: confusion matrix of test 1 result

```

--- Model Result ---

Confusion Matrix:
[[8545 4825]
 [1246 3226]]

Classification Report:

```

	precision	recall	f1-score	support
Not Canceled	0.87	0.64	0.74	13370
Canceled	0.40	0.72	0.52	4472
accuracy			0.66	17842
macro avg	0.64	0.68	0.63	17842
weighted avg	0.75	0.66	0.68	17842

```

Accuracy: 65.97%
AUC-ROC Score: 0.7311

```

Figure 5.3: More detailed information on test 1 result

Figures 5.2 and 5.3 show the confusion matrix and detailed information for the current Test 1 result. We can see that the accuracy is 66%, which is moderate. Looking at the F1-score, this model performs better at predicting customers who will not cancel their bookings ($0.74 > 0.52$). With this in mind, we can move on to the next test.

X features fit more GaussianNB (second test)

For the second test, I used variables that are more appropriate for the GaussianNB model. These variables are:

- Lead_time

- Arrival_date_week_number
- Arrival_date_day_of_month
- Adr

This time, the variable selection is significantly reduced, and I believe these variables are more suitable for GaussianNB. Let's examine the results:

```

--- Model Result ---

Confusion Matrix:
[[12764   606]
 [ 4096   376]]

Classification Report:

```

	precision	recall	f1-score	support
Not Canceled	0.76	0.95	0.84	13370
Canceled	0.38	0.08	0.14	4472
accuracy			0.74	17842
macro avg	0.57	0.52	0.49	17842
weighted avg	0.66	0.74	0.67	17842

```

Accuracy: 73.65%
AUC-ROC Score: 0.6407

```

Figure 5.4: detail information of test 2 result

This model (Figure 5.4) performs significantly worse than the Test 1 model, with an AUC of only 0.6407, lower than the first one (0.73). Although the accuracy increased and it is even better at classifying customers who "Not canceled," this model is terrible at classifying the cancellation

rate. Therefore, this model is essentially unusable for a hotel that wants to predict whether a customer will cancel their booking. In this situation, I believe the focus should be on identifying those who canceled their booking so that informed decisions can be made to address the loss of customers.

Therefore, the Test 1 model is still the best performing model so far. It seems to indicate that including all available features might not always be detrimental..

MultinomialNB (third test)

For the third test, I decided to use MultinomialNB, focusing only on the discrete count variables.

Here is the list of variables I'm using for this test:

- stays_in_weekend_nights
- stays_in_week_nights
- adults
- children
- babies
- previous_cancellations
- previous_bookings_not_canceled
- booking_changes
- days_in_waiting_list
- total_of_special_requests

Here is the result:

```
--- Model Result ---

Confusion Matrix:
[[13186   184]
 [ 4234   238]]

Classification Report:
              precision    recall  f1-score   support

Not Canceled    0.76      0.99      0.86    13370
   Canceled    0.56      0.05      0.10     4472

   accuracy              0.75    17842
  macro avg    0.66      0.52      0.48    17842
weighted avg    0.71      0.75      0.67    17842

Accuracy: 75.24%
AUC-ROC Score: 0.6550
```

Figure 5.5: detailed information of test 3 result

From Figures 5.4 and 5.5, it's evident that both smaller models share the same issue: they perform worse at predicting cancellations, which is undesirable. Although I believe this model is slightly better than the Test 2 model, it still cannot outperform the Test 1 model simply because its AUC is lower and it is less effective at classifying cancellations.

BernoulliNB Dumping everything(fourth test)

In this test, I used the same variables as in Test 1, which means including every possible variable in the model. The rationale behind this is that a large proportion of the variables in this hotel

dataset are categorical, so I one-hot encoded every categorical variable. I also included the Country and Agent columns (as dummy variables) because I believe they are related to the cancellation rate. Here is the performance:

```
--- Model Result ---

Confusion Matrix:
[[9982 3388]
 [1378 3094]]

Classification Report:
              precision    recall  f1-score   support

Not Canceled    0.88      0.75      0.81     13370
   Canceled    0.48      0.69      0.56      4472

   accuracy              0.73     17842
  macro avg    0.68      0.72      0.69     17842
weighted avg    0.78      0.73      0.75     17842

Accuracy: 73.29%
AUC-ROC Score: 0.7828
```

Figure: 5.6: detailed information of test 4 result

Figure 5.6 shows very promising results! By switching to BernoulliNB, the performance surpassed the Test 1 model with an AUC of 0.78 compared to 0.73. Not only that, but the accuracy is also significantly better than Test 1 (73.29% vs. 65.97%). The F1-score for cancellation improved by 0.04, which is a great improvement. This model is now better than the original Test 1 model.

Selective BernoulliNB (fifth test)

In this final test, I wanted to explore what would happen if I only included the categorical dummy variables. Therefore, I dropped columns like lead_time, adr, and other discrete and continuous variables. Here is the performance:

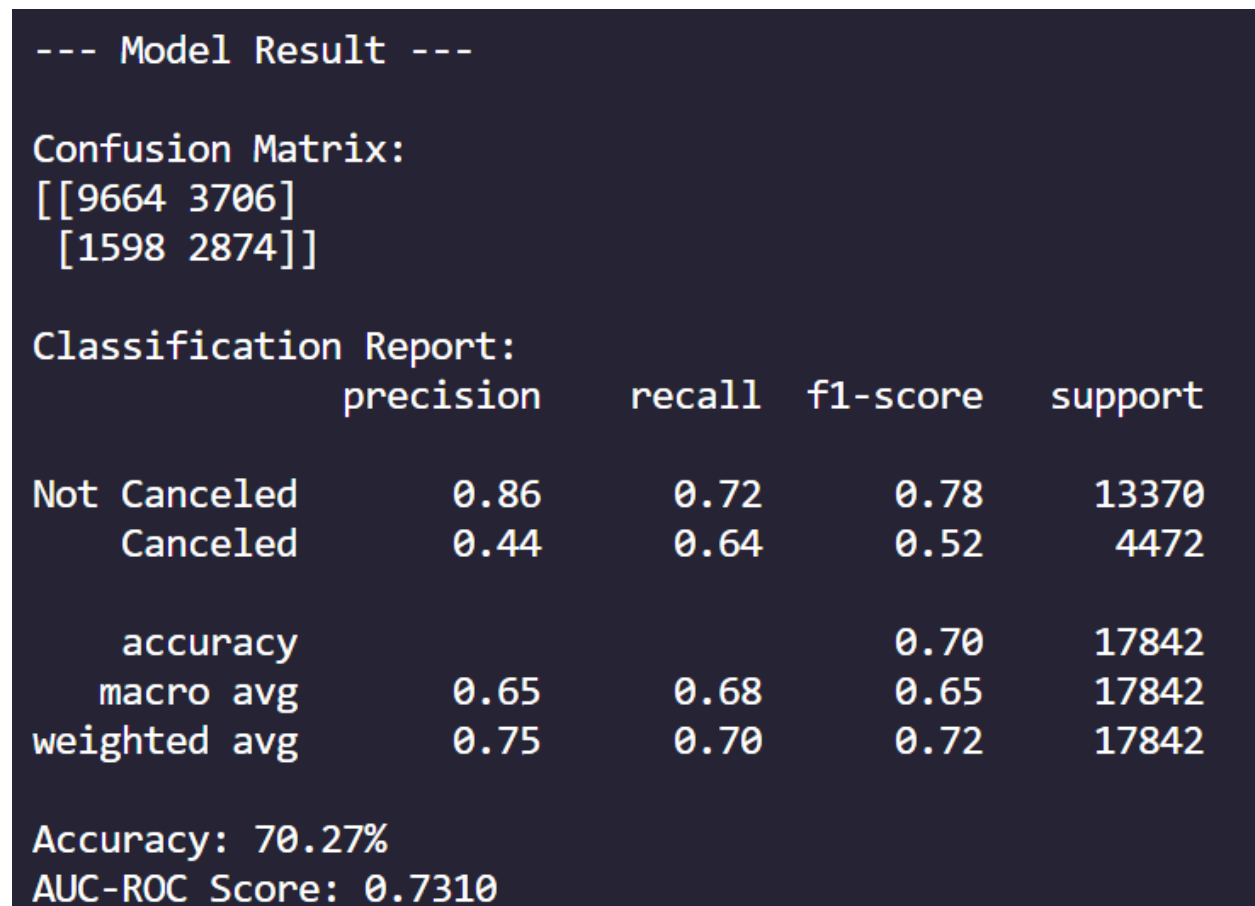


Figure 5.7 detailed information of test 5 result

As we can see in Figure 5.7, the result is actually worse than Test 4. After running Tests 5, 2, and 3, I noticed that providing more information to the model in the hotel dataset sometimes allows NB to perform better, even though the additional variables might not be ideal.

Conclusion

The best model throughout the tests is BernoulliNB with a mixture of variables (meaning using all available information). This model has the highest AUC at 0.78, and the highest F1-score for cancellation at 0.56.

4.2 Decision Trees

In this section, I implement a decision tree model to classify customer hotel booking cancellations. The key objective here is to identify the best-performing decision tree model based on AUC, F1-score for classifying cancellations, and overall accuracy.

Methodology

For this model, I have 71,000 records and 577 columns due to the dummy variables created for the country and agent columns. A notable advantage of the decision tree model compared to the previous NB model is its insensitivity to the mixture of variables. I split the data into 75% for training and 25% for testing, resulting in a training set size of 53,526 and a testing set size of 17,842. The Gini index is used as the algorithm for determining the branches. Furthermore, I primarily implement pre-pruning by adjusting the maximum depth, a crucial method to help me identify the best model.

Findings

To find the optimal model, I incrementally adjusted the maximum depth until the model's performance began to decline. This indicates the point at which the branches become overly complex and the model starts to overfit. After several tests, ranging from `max_depth = 3` to

`max\_depth = 15`, the AUC began to drop after 13. This suggests that setting `max\_depth = 13` yields approximately the best decision tree model. The results are as follows:

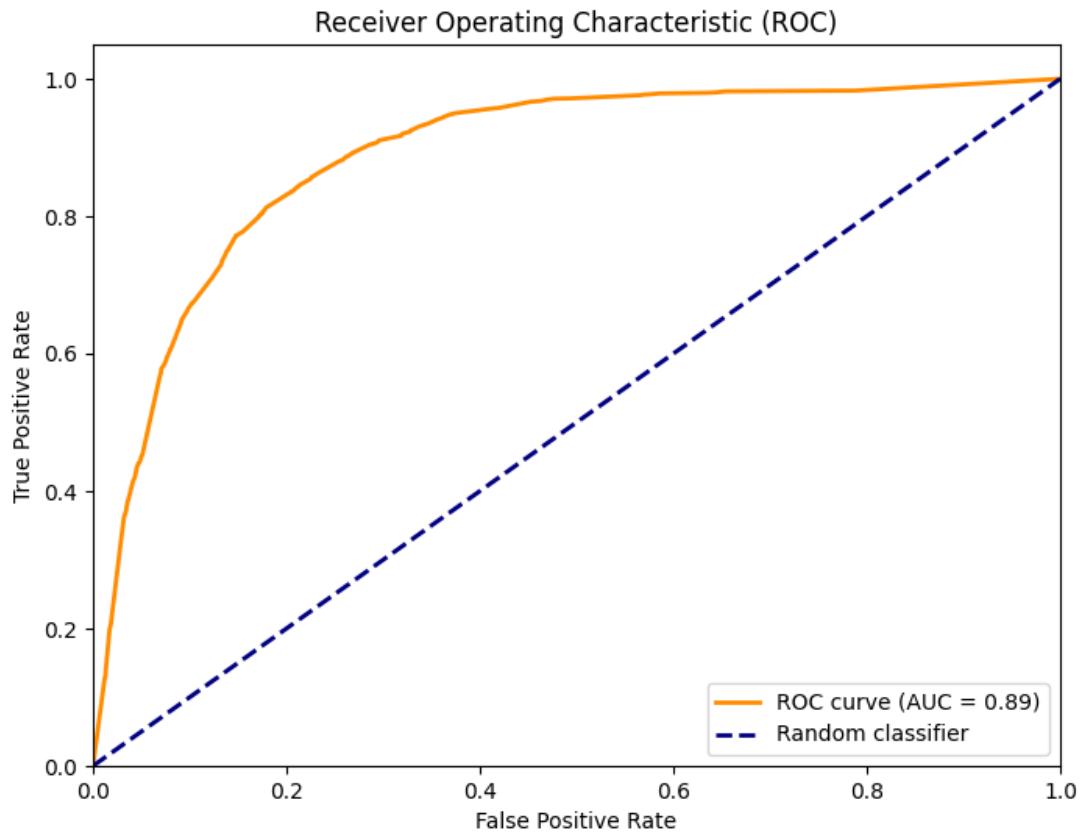


Figure 6.1: AUC-ROC for optimal decision trees model

As shown in Figure 6.1, the AUC is approximately 0.89, which I believe represents the best model for classifying hotel cancellations thus far.

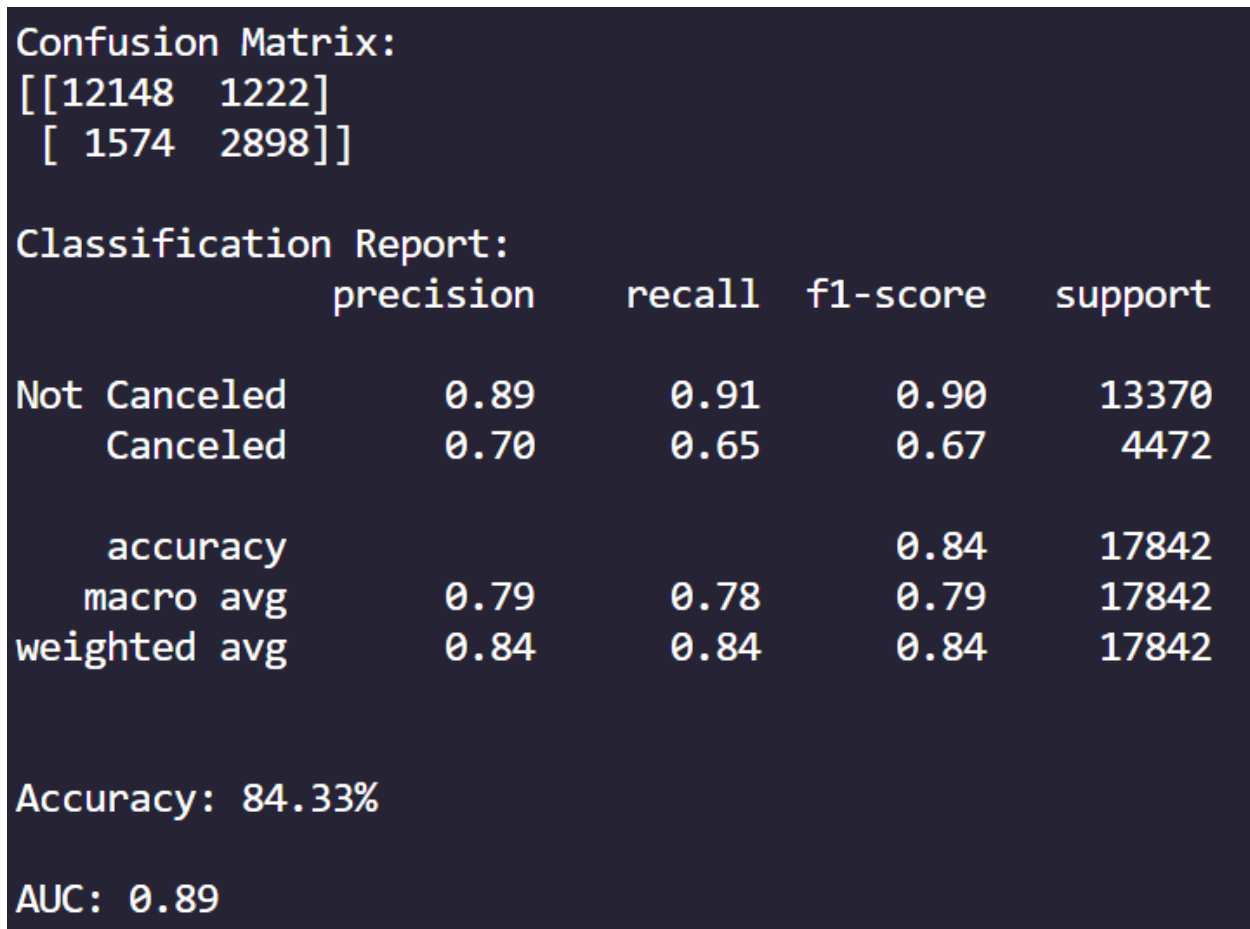


Figure 6.2: detailed statistics for decision trees model

Figure 6.2 shows an accuracy of 84.33%, which is also the highest accuracy achieved so far.

Based on the F1-score, this model performs best at classifying customers who have not canceled.

However, my primary goal is to focus on customers who do cancel their bookings. The F1-score of 0.67, which indicates a moderately high performance, further supports the conclusion that this is currently the best-performing classification model.

Conclusion

With an AUC of 0.89, this model is currently the best performing in terms of classification. I now need to focus on the final model, SVM, and then compare its overall performance in the subsequent model comparison section.

4.3 Support Vector Machines

This section focuses on applying the Support Vector Machine (SVM) model to the hotel dataset.

Methodology

I will use SVC, a Support Vector Machine methodology, for clustering with the goal of classifying customer cancellations. A known drawback of SVM is its high CPU usage; my cloud service can only handle 1000 records. In this model, I will also apply 577 columns of dummy variables to match the decision trees model as closely as possible, and the test sample will consist of 25% of the 1000 records.

Findings

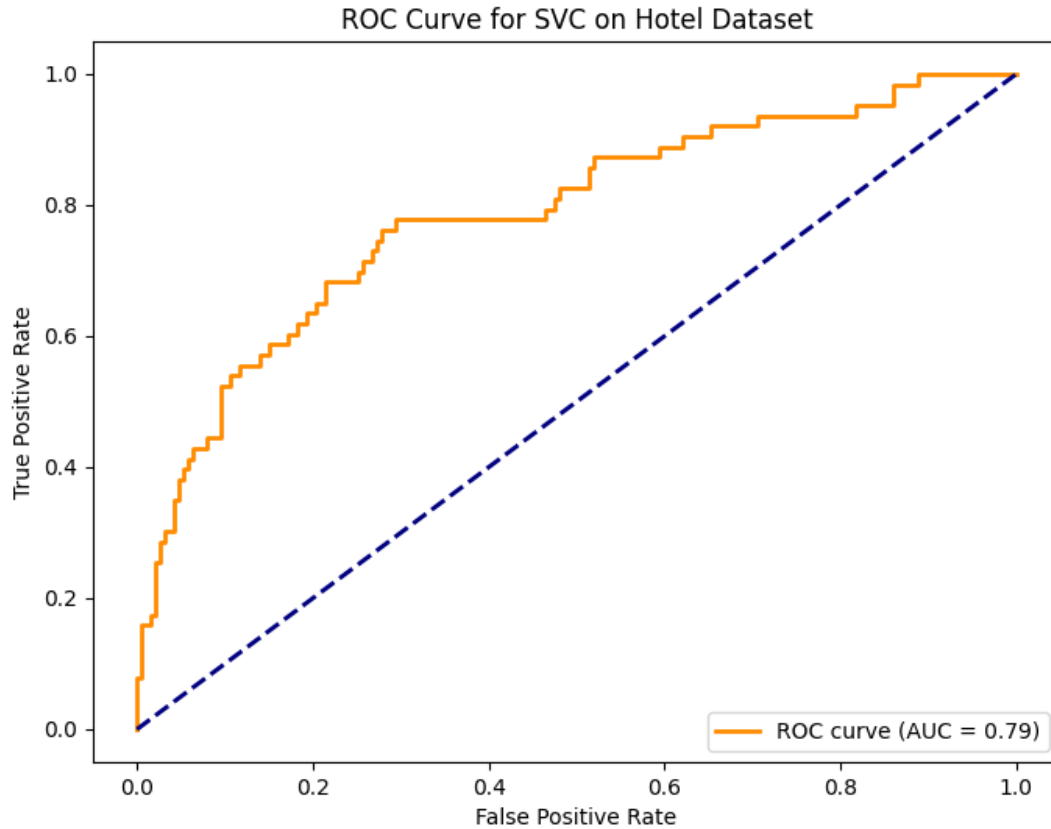


Figure 7.1: AUC-ROC result for SVC model

The AUC of 0.79 shown in Figure 7.1 is quite good, especially considering that I only used 1000 records from the original dataset of 71,000. I was initially concerned about potential zero-division errors during computation, as this hotel dataset contains many discrete and binary variables, and using only 1000 samples increases the likelihood of encountering such errors.

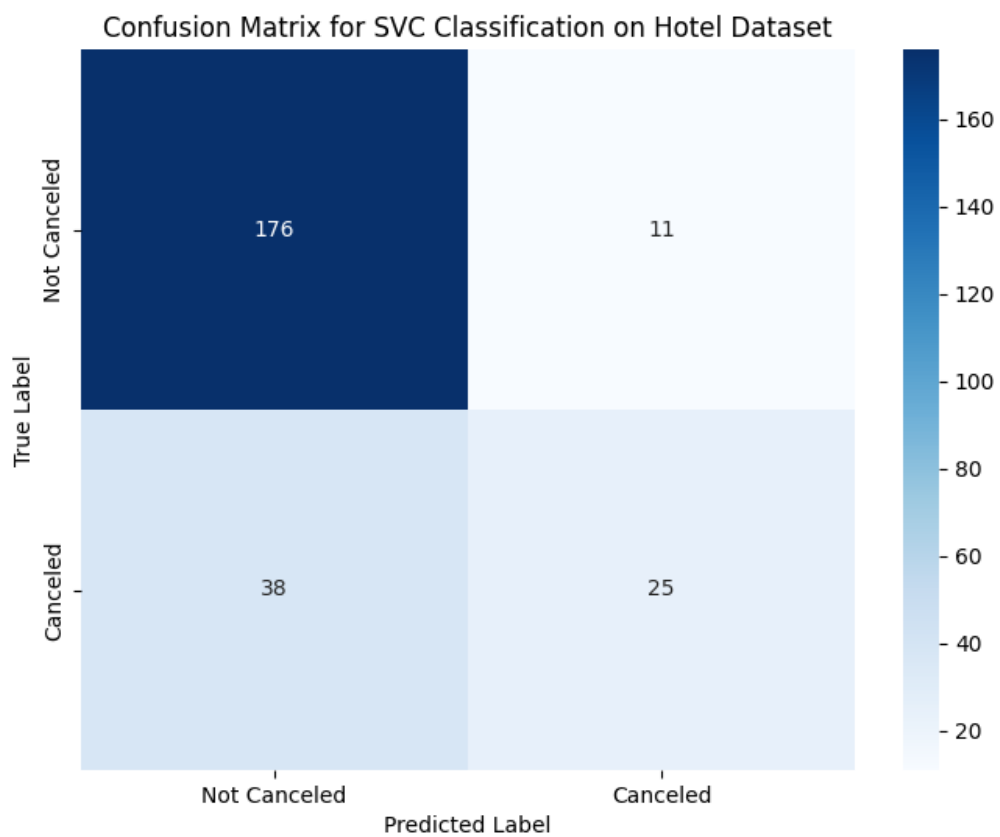


Figure 7.2: confusion matrix of SVC model

```

Training SVC model on Hotel Dataset...
Best Model Parameters: {'C': 8, 'gamma': 'auto', 'kernel': 'rbf'}
Model does not provide coefficients (likely due to kernel type).
SVC Classification on Hotel Dataset Accuracy: 0.8040
Classification Report:

```

	precision	recall	f1-score	support
Not Canceled	0.82	0.94	0.88	187
Canceled	0.69	0.40	0.51	63
accuracy			0.80	250
macro avg	0.76	0.67	0.69	250
weighted avg	0.79	0.80	0.78	250

```

SVC Classification on Hotel Dataset ROC AUC Score: 0.7861

```

Figure 7.3: detailed statistical information of MVC model

Figures 7.2 and 7.3 show the results of this model. After testing several parameter combinations, the best results were achieved with $C=8$, $\gamma='auto'$, and $\text{kernel}='rbf'$. While this model performs reasonably well in terms of accuracy and AUC and is not the worst I've encountered, the F1-score for classifying cancellations is 0.51. This is moderate, but insufficient for reliably predicting new cancellations, as the prediction is essentially a 50/50 chance.

Conclusion

Theoretically, SVC has the potential to perform very well, and I am eager to test it on the full dataset. Using only 1000 records feels inadequate, especially when compared to other models. I believe that with the full dataset and optimized parameters, SVC could become a robust model for this hotel dataset. However, a significant drawback is its high computational cost. Budget-

conscious businesses may prefer less computationally expensive, albeit potentially less robust, models, which would make SVC difficult to implement.

4.4 Comparative Analysis

In this section, I will compare the three models—Naïve Bayesian, Decision Trees, and SVC—side-by-side to select the best model for predicting which customers will cancel their booking at the Portugal hotel. Here is the comparison table:

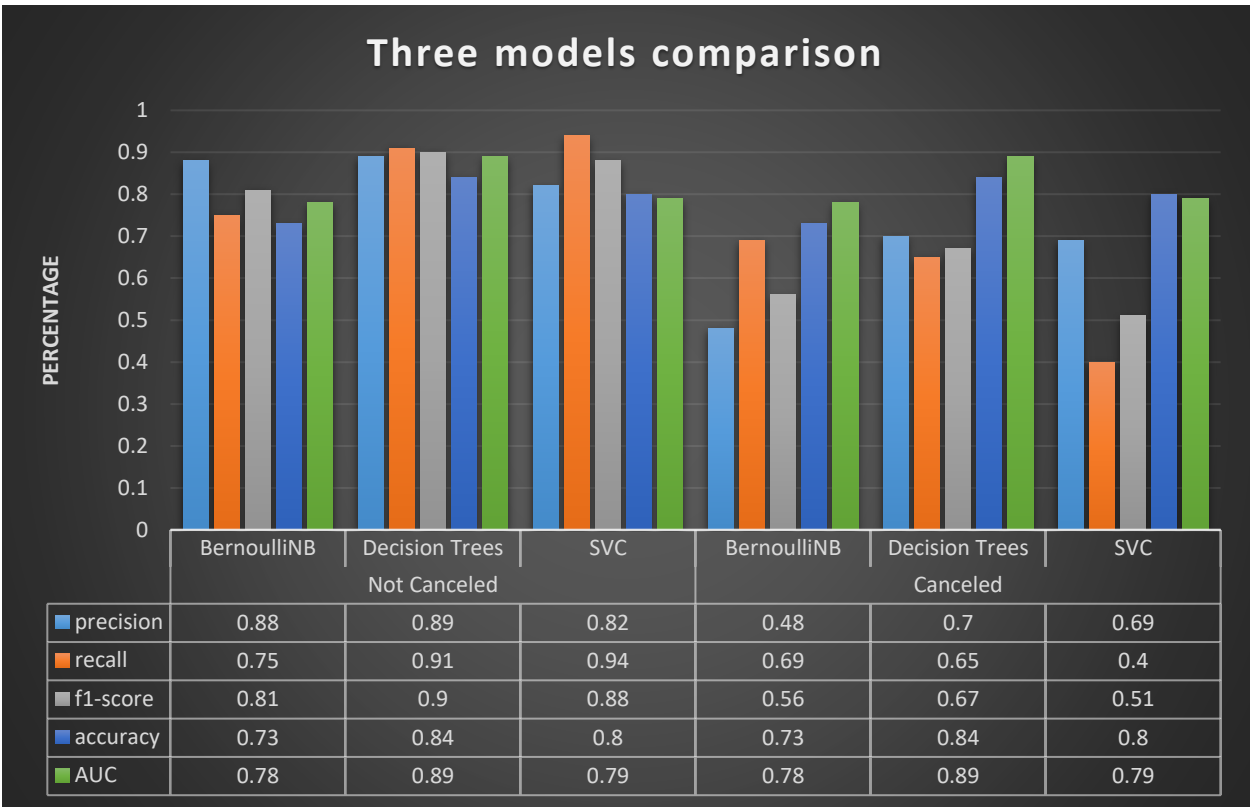


Figure 8: model comparison for all three models

Overall comparison

In this section, I will compare these three models based on their overall accuracy and AUC, as these metrics are crucial for evaluating general model performance. In Figure 8, decision trees

have the highest accuracy (0.84) compared to the other models. This indicates that after training and using the model to predict the 25% of the testing data, the decision tree model had an 84% correction rate in predicting both cancellation and non-cancellation customers, which is quite good considering the inherent difficulty in predicting customer cancellations.

It's worth noting that although the decision tree model performs best in terms of accuracy, the SVC model is very close behind (0.8). Additionally, the decision tree used the full dataset of 71,000 records, while SVC used only a sample of 1,000 records. Therefore, it's important to remember that SVC could potentially be more powerful than the decision tree model if tested on the full dataset. The more data SVC learns from, the more accurately it can identify the optimal boundary for separating cancellation and non-cancellation customers. So, even though SVC doesn't win this round, its performance is still surprisingly good.

In terms of area under the curve (AUC), the decision tree model dominates (0.89). This means the decision tree model excels at separating cancellation and non-cancellation customers. Combined with its accuracy performance, this makes it the best model overall for classifying cancellations.

Detailed comparison

In this section, I will primarily focus on a detailed comparison of the three models. The reason for this is that the hotel is likely less concerned about customers who will not cancel their bookings, as it's generally assumed that those customers will proceed with their reservations. The hotel is more focused on customers who cancel their bookings, as they are the ones who cause the hotel to lose revenue. Therefore, I need to focus on predicting customer cancellations to effectively address this key business problem.

All three models perform well in predicting non-cancellation customers, with F1-scores all higher than 0.8, which is very good. Still, decision trees and SVC have very similar F1-score performance (0.9 vs. 0.88). Notably, SVC has the highest recall (0.94) among the three models, meaning that the SVC model excels at identifying 'real' non-cancellation customers, making it more trustworthy than the other two models in this regard. However, it has a slightly lower precision, which impacts its F1-score. Consequently, the decision tree model remains the best overall.

In terms of predicting customers who cancel their bookings, the decision tree model's F1-score dominates the other two models (0.67), while SVC performs significantly worse. The poor cancellation prediction performance of SVC further supports my point that SVC's performance heavily relies on the size of the data sample. It performs well in identifying non-cancellation customers because the majority of customers do not cancel their bookings. However, when it comes to predicting cancellations, the sample likely has significantly fewer examples of cancellations.

Conclusion

The decision tree model has emerged as the best model across all comparisons. Here's how I would interpret its performance and implement it at the Portugal hotel: When a new customer books a room through the hotel's online app, the hotel can use this model to predict whether the customer will cancel the booking based on the information received. There are two possible scenarios: First, if the model predicts that the customer will not cancel, it indicates to the hotel manager that, 90% of the time, this customer will likely not cancel their booking. Second, if the model predicts that the customer will cancel, it suggests to the hotel manager that, 67% of the time, this customer is likely to cancel their booking, so the hotel may need to develop a strategy to retain this customer. In this way, the hotel can save money by not investing resources in retaining non-cancellation

customers, and instead focus its efforts and additional costs on retaining customers who are likely to cancel.

5. Association Rules Analysis

This section primarily focuses on association rule mining. The key objective here is to identify patterns in the combinations of choices customers make when booking hotels. This will enable me to help hotels in Portugal gain valuable insights and develop informed marketing and operational strategies based on my findings.

For the association rule mining, I prepared these variables for the model:

- Meal_plans
- Room_types
- Week_numbers
- Customer_type
- Market_segments
- Required_parking_spaces
- Total_of_special_requests
- Children_count
- Adults_count
- Stays_in_weekend_nights
- Stays_in_week_night

The reason I'm including these variables is to uncover the preferences of different customer segments. Since diverse customers have varying tastes, providing personalized hotel recommendations can be challenging. I believe these variables will help the hotel identify patterns that can lead to better-informed decisions.

Methodology

I used the full-sized dataset (over 71,000 records) to run this association model, and the performance has been quite good. The algorithm I'm using is the Apriori algorithm, a traditional approach that helps identify frequent itemsets for the rules. While not as efficient as the FP-Growth algorithm, it has proven sufficient for handling 71,000 records in this case.

For all the variables mentioned earlier, I created a separate CSV file specifically for this model because all the variables need to be one-hot-encoded (converted into dummy variables). The nature of the hotel dataset makes it relatively easy to convert all columns to corresponding dummy variables, as virtually all variables are discrete or categorical, and the range for the discrete variables is moderate. This contributes to the strong model performance.

Two key settings in the model are the minimum support threshold for the frequent itemsets and the minimum confidence threshold for the rules. I chose a minimum support threshold of 0.2, which I believe is suitable for the current dataset. I could potentially go lower due to the dataset size, but 0.2 seems a reasonable starting point for this section. For the minimum confidence threshold, I set it to 0.6. This ensures that when interpreting the results, I can be reasonably confident (over 50%) that there are associations between the two groups of itemsets. To be more realistic, I believe any confidence level above 0.6 is acceptable for discussion with the hotel. However, for other businesses, the appropriate threshold may vary.

Findings

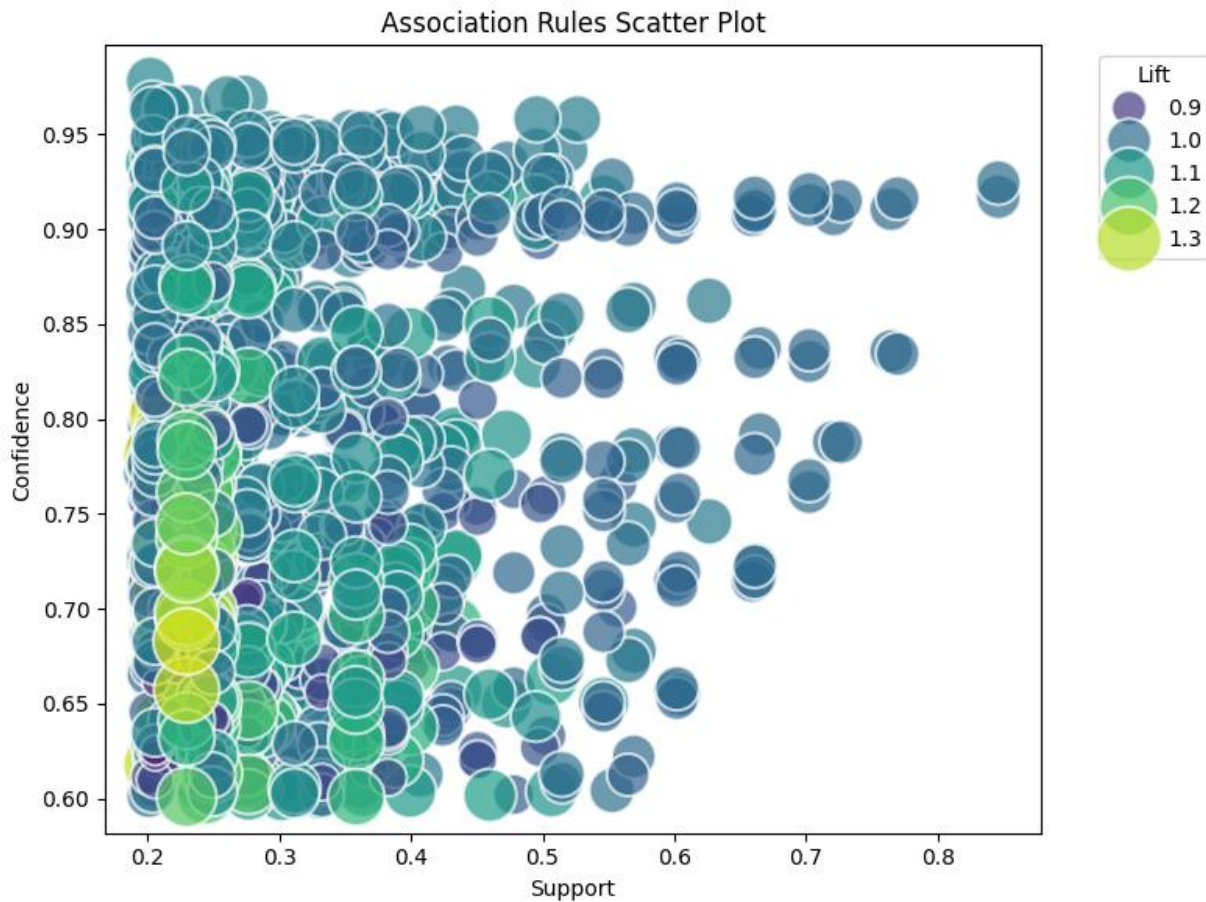


Figure 9.1: scatter plot of rules lift distribution

Here is the scatter plot (Figure 9.1) for the current model, which contains approximately 1113 total rules. Interestingly, the higher-lift rules tend to cluster in the bottom left corner of the graph, indicating a support range of 0.2 to 0.3 and a confidence level between 0.6 and 0.8. This characteristic suggests that I could potentially lower the minimum support threshold (MST) to uncover additional potentially high-lift rules. However, solely focusing on high-lift associations can be misleading, as a low conviction score (<1) for a particular high-lift rule indicates a lack of reliability.

antecedents	consequents	support	confiden ce	lift	convicti on
customer_type_Transi ent, total_of_special_reque sts_1	children_0.0, market_segment_Onli ne TA	0.215432 687	0.735645 933	1.360567 507	1.737478 45
customer_type_Transi ent, required_car_parking_ spaces_0, total_of_special_reque sts_1	market_segment_Onli ne TA	0.211803 609	0.802207 716	1.343974 279	2.038036 207
children_0.0, customer_type_Transi ent, total_of_special_reque sts_1	market_segment_Onli ne TA	0.215432 687	0.794737 93	1.331459 814	1.963869 097
assigned_room_type_ A, market_segment_Onli ne TA	adults_2, children_0.0, customer_type_Transi ent,	0.229850 914	0.682788 762	1.326688 493	1.530032 712

	required_car_parking_spaces_0				
customer_type_Transient, total_of_special_requests_1	market_segment_Online TA	0.231588 387	0.790813 397	1.324884 869	1.927025 105
required_car_parking_spaces_0, total_of_special_requests_1	children_0.0, market_segment_Online TA	0.214423 831	0.709523 368	1.312254 165	1.581227 003
adults_2, assigned_room_type_A, customer_type_Transient	children_0.0, market_segment_Online TA, required_car_parking_spaces_0	0.229850 914	0.657211 538	1.310200 65	1.453924 766
customer_type_Transient, total_of_special_requests_1	market_segment_Online TA, required_car_parking_spaces_0	0.211803 609	0.723253 589	1.308552 505	1.616235 229

adults_2, total_of_special_reque sts_1	market_segment_Onli ne TA	0.206549 154	0.781062 894	1.308549 417	1.841203 819
--	------------------------------	-----------------	-----------------	-----------------	-----------------

Table 1: top 10 high lift result of Association rule model

Table 1 displays the top 10 results, sorted by lift in descending order. These rules exhibit both high lift and high conviction, making them highly reliable. Notably, almost all the rules include the "Online TA" market segment. This suggests that these antecedents frequently lead to customers booking the hotel through an online travel agent. Consequently, if we fulfill or encounter any of the antecedents in these top 10 results, we can be over 60% confident that customers will book through an online travel agent.

For example, consider the first result: if a customer is a "Transient" (meaning they are staying at the hotel for a very short period) and has a special request (such as requesting flowers and a cake on the bed for a birthday), there is a 73% chance they will not bring any children and will also book through an online travel agent. Many other rules offer valuable insights, and we will explore further insights and business strategies in later sections.

Conclusion

The association analysis has proven effective in this section, allowing me to extract significant insights from the results. The strategies analysis will be presented in subsequent sections.

6. Clustering Analysis

This section employs hierarchical clustering to identify potential customer segments within a hotel dataset comprising 5000 samples and 9 features. The objective is to evaluate the model's performance and analyze the resulting clusters for actionable insights.

Methodology

Agglomerative clustering, a bottom-up approach, was implemented on the dataset. While the sample size may present a limitation, potentially impacting the granularity of results (and suggesting that further data acquisition could enhance the analysis), the model incorporates the following features:

- lead_time
- adr
- required_car_parking_spaces
- total_of_special_requests
- is_city_hotel
- adults
- children
- is_repeated_guest
- is_canceled

The clustering model utilizes Ward linkage with Euclidean affinity. This combination is well-suited for handling the dimensionality of the data because Ward linkage is effective at creating

compact clusters by minimizing variance, which helps when dealing with many features. At the same time, Euclidean affinity provides a straightforward and commonly used way to measure distance between data points in a multi-dimensional space, making it a robust choice for datasets with many features.

Findings

To determine the optimal number of clusters, a dendrogram was generated (see figure below). This visualization serves as a key tool in identifying the appropriate granularity for customer segmentation.

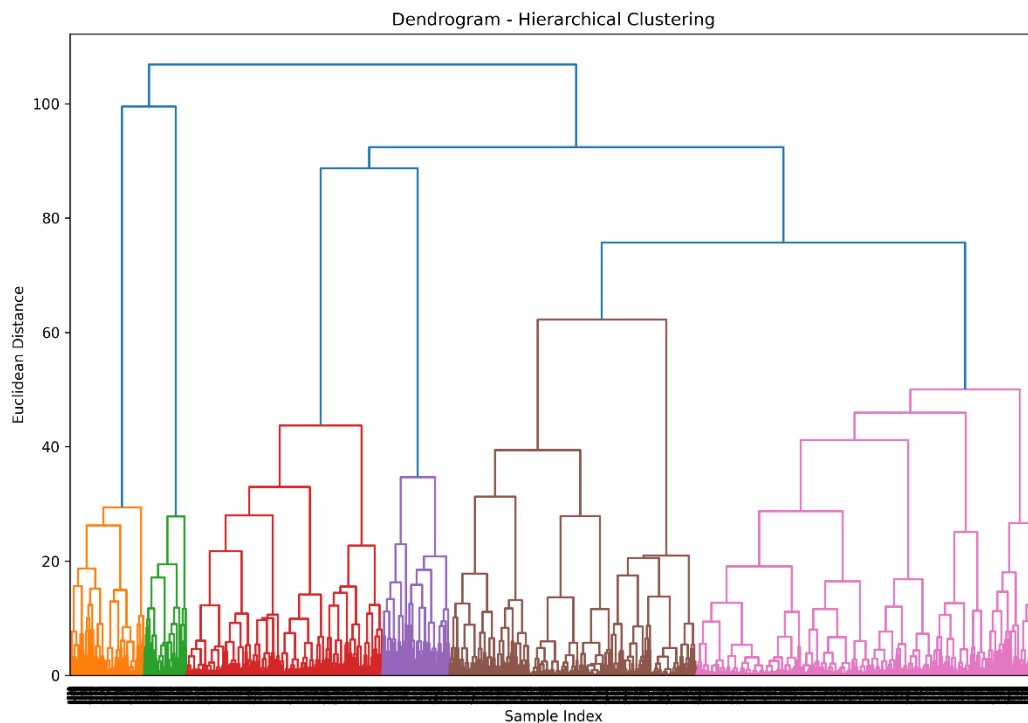


Figure 10.1: dendrogram shows 7 clusters

The dendrogram (figure 10.1) suggests seven distinct clusters (represented by seven colors) within the selected features. This informed the final number of clusters used in the model.

The model's performance is detailed below:

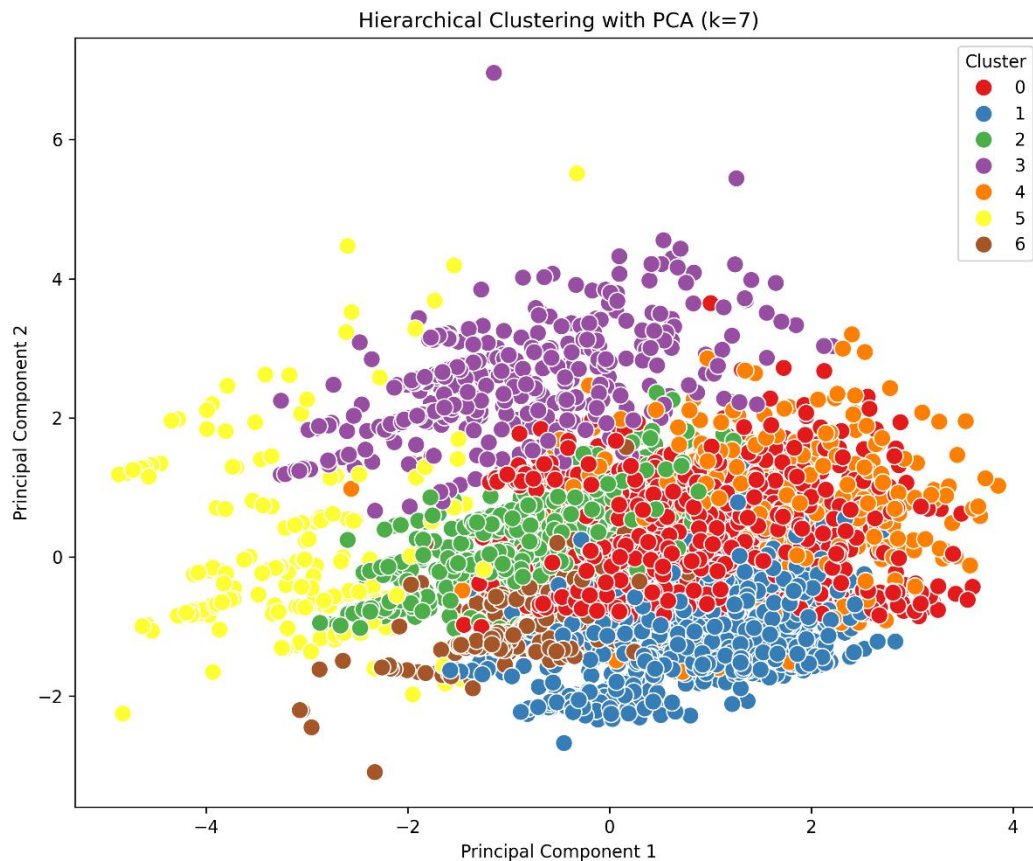


Figure 10.2: PCA is 2-dimensional space for clustering

The PCA visualization shows some well-defined clusters (e.g., cluster 5), while others exhibit overlap. This overlap suggests that the Silhouette Score might be relatively low.

```
Clustering completed with 7 clusters
Silhouette Score: 0.2309
```

Figure 10.3: Silhouette Score for model evaluation

Indeed, the Silhouette score of 0.2309 (figure 10.3) is close to zero. This is somewhat expected, given that we are using the original dataset (where sampling bias might affect the score) and that distinguishing customers based on booking habits in real-world data is inherently difficult. Therefore, this score range is not unexpected.

A brief cluster profile follows:

Cluster Profiles (Mean Values):								
	0	1	2	3	4	5	6	
size	1762.000000	1014.000000	836.000000	382.000000	347.000000	219.000000	440.000000	
percentage	35.240000	20.280000	16.720000	7.640000	6.940000	4.380000	8.800000	
lead_time	60.941544	74.012821	45.674641	41.243455	69.832853	8.173516	36.693182	
adr	107.395783	104.384615	78.939282	99.725105	135.583429	64.830594	90.795023	
required_car_parking_spaces	0.000000	0.000000	0.000000	1.007853	0.002882	0.164384	0.000000	
total_of_special_requests	0.875142	0.491124	0.478469	0.824607	0.899135	0.698630	0.425000	
is_city_hotel	0.933031	0.708087	0.000000	0.269634	0.714697	0.493151	0.997727	
adults	2.148127	1.828402	1.754785	1.937173	1.902017	1.347032	0.988636	
children	0.000000	0.000000	0.000000	0.136126	1.371758	0.022831	0.000000	
is_repeated_guest	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000	0.000000	
is_canceled	0.054484	1.000000	0.000000	0.000000	0.371758	0.045662	0.009091	

Figure 10.4: model features insides (mean)

This table (figure 10.4) unveils seven distinct groups of hotel customers, each with unique preferences and behaviors.

Cluster 0:

Representing the largest segment at 35.24% of customers, Cluster 0 appears to be families or couples favoring city hotels. The average cluster size is the largest at 1762, indicating it's indeed the most populated. They exhibit a moderate lead_time of approximately 60.9 days before booking. Notably, they are willing to pay a higher adr of around 107.4, suggesting a value for city hotel convenience. Furthermore, the "is_city_hotel" value of 0.93 confirms a high probability of city hotel preference within this group. With an average of 2.15 adults per booking, this supports the idea of families or couples. This cluster presents a significant

opportunity for city hotels to target with tailored marketing, potentially using dynamic pricing to leverage their willingness to pay a higher ADR while offering value for longer stays or off-peak periods.

Cluster 1:

Comprising 20.28% of customers, Cluster 1 shows a strong tendency towards cancellation, indicated by an "is_canceled" value of 1.00. While this might be a mean value implying a very high likelihood rather than absolute 100%, it highlights a critical characteristic. This cluster also books with the longest lead_time at approximately 74 days. The average cluster size is 1014. The adr is around 104.4. The combination of high cancellation and long lead time suggests these bookings are made far in advance and are more speculative. Hotels should investigate cancellation reasons and consider offering flexible booking options or enhancing pre-arrival engagement to mitigate cancellations within this segment.

Cluster 2:

This segment constitutes 16.72% of customers, with a cluster size of 836. A key feature is their lower preference for city hotels, as indicated by an "is_city_hotel" value of 0.00. Coupled with a lower adr of around 78.9 and a moderate lead_time of 45.7 days, it points to a value-conscious segment seeking non-city hotel options. They also show 0.00 for "required_car_parking_spaces", which is interesting but might not be a defining characteristic. The average number of adults is around 1.75. This cluster represents a target for hotels outside city centers, where marketing should emphasize value, tranquil environments, and amenities of resort or suburban locations.

Cluster 3:

Representing 7.64% of customers (cluster size of 382), Cluster 3 is distinguished by a high need for car parking, with a "required_car_parking_spaces" value of 1.01. This strongly suggests travelers arriving by car. They also exhibit a relatively high number of "total_of_special_requests" at 0.82. The lead_time is moderate at 41.2 days, and the adr is around 99.7. The average number of adults is approximately 1.94. Hotels with ample parking and good special request handling should target this segment, highlighting parking availability and responsiveness to guest needs in their marketing.

Cluster 4:

A smaller segment at 6.94% (cluster size 347), Cluster 4 stands out as the highest-spending group with an adr of around 135.6. They also have a notable average of 1.37 children per booking, the highest among all clusters. The lead_time is longer at 69.8 days. However, they also have a moderate cancellation rate, with "is_canceled" at 0.37. The number of "total_of_special_requests" is also relatively high at 0.90. This indicates affluent families willing to pay a premium, but with higher expectations and potentially more complex planning. Premium family-focused hotels should target this segment, emphasizing luxury amenities and exceptional service, while considering flexible cancellation options.

Cluster 5:

One of the smallest at 4.38% (cluster size 219), Cluster 5 is defined by loyalty, showing an "is_repeated_guest" value of 1.00. They also have the lowest adr at around 64.8 and the shortest lead_time at just 8.2 days. The cancellation rate is low, with "is_canceled" at 0.05. The average number of adults is 1.35. This strongly suggests price-sensitive, frequent travelers, likely

business travelers valuing efficiency and cost-effectiveness. Loyalty programs are ideal for this segment, along with streamlined booking and business-relevant amenities.

Cluster 6:

Representing 8.8% of customers (cluster size 440), Cluster 6 appears to be solo city explorers. They have the highest "is_city_hotel" value at 0.997, practically exclusive city hotel preference. The average number of adults is very low at 0.99, essentially single occupancy. The cancellation rate is very low, "is_canceled" at 0.009. The lead_time is moderate at 36.7 days, and the adr is around 98.8. This points to reliable solo travelers preferring city hotels. Marketing should target solo travelers with city-centric offers, focusing on location and safety, and potentially rewarding their high booking reliability.

Conclusion

The Agglomerative clustering results are moderate and open to interpretation, highlighting the inherent trade-offs between different models. Sample size appears to be a limitation; my experiments suggest that reducing the sample size of the hotel dataset negatively impacts model performance. Therefore, based on my experience with these labs, increased computational power can, in some cases (but not always), lead to improved results.

7. Discussion & Recommendations

7.1 Business recommendations

Business recommendation through OLS-HAC model

Here are several recommendations for the business owner of the hotel to improve their strategies for hotel room price:

1. **To avoid longer lead times:** According to the model (Figure 11.1), an additional unit of lead time will cause a room price drop of \$0.12. Longer lead times also increase the chance of booking cancellations (as shown in the clustering model), which is why we need to avoid them. One strategy hotels can use is to provide special discounts and opportunities for customers to get lower prices if they arrive a little earlier.
2. **Introduce hotel booking combos:** Each child and adult increases the ADR by approximately \$13, which is expected because more people often require larger rooms, leading to more revenue. Therefore, marketing segment discounts to family customers could greatly increase the hotel's revenue.
3. **Consider expanding the parking space:** For each customer who requires a parking space, the ADR increases by \$6. This is likely because customers who require parking spaces are often repeat customers or consistent travelers. Since they also pay for parking, this creates another revenue stream to increase overall revenue.
4. **Social media marketing for city hotels:** If you own a city hotel, try marketing the local places to eat and entertainment options available to your customers. City hotels tend to have prices \$28 higher than other types of hotels, so any marketing strategies that incentivize customers to stay in a city hotel are highly beneficial.
5. **Meal plan recommendations:** Introduce appealing meal plans and showcase high-quality photographs of the food offered at the hotel. Meal plans are very profitable and represent an important revenue stream for the hotel.
6. **Social media post in Summertime:** During the summer, the ADR tends to increase due to more people being on vacation. Therefore, implementing marketing strategies such as

influencer incentives and email campaigns targeting existing customers can significantly boost revenue.

Business recommendations based on decision trees model

Given the best model (decision trees), I have several recommendations for the hotel manager when they are implementing this model.

- **Resource allocation and rearrangement:** Although hotels primarily focus on customers who cancel their bookings, the model is remarkably effective at predicting customers who will not cancel. By leveraging this capability, we can reallocate excess resources, such as communication efforts involving frequent phone calls or text messages, to avoid unnecessary costs. Instead, we can dedicate more resources to addressing potential cancellations, thereby optimizing the utilization of management resources.
- **Booking contract reinforcement:** The hotel can reinforce its terms of service to prevent unreasonable cancellations. For example, requiring pre-authorization on a credit card and holding a certain amount to secure the room ensures that customers understand the importance of their booking selection. If the customer cancels, they would not receive a refund of the held amount. This method ensures the customer understands the importance of their booking and, even if they cancel, the hotel recoups some money as a penalty.
- **Enhance customer communication:** For customers likely to cancel their bookings, increased communication is crucial to securing the reservation. Many cancellations stem from miscommunication regarding the hotel booking process. By combining reinforced communication with the resource allocation optimization strategy mentioned earlier, the hotel can secure a larger number of bookings and further improve revenue.

Business recommendation based on association rules mining

- **Social media couple marketing with hotel online link offering combo for room type**

A: The rationale behind this strategy is that, according to association rule mining, 68% of bookings for two adults are for room type A. This suggests that room A is popular among couples. Since they are more likely to be drawn to marketing for room type A due to their preferences when using online social media and booking platforms, this strategy will be particularly effective during the summer. Previous OLSHAC results have shown that more couples book hotels during the summer months.

- **Room type D plus meal plan bed and breakfast:** Room D has a strong association with the bed & breakfast meal plan, with 78% confidence. Therefore, to target customers who choose room type D, we can easily recommend this meal plan to enhance communication and potentially increase sales.
- **If paying commercial for children related segment, STOP!:** The rationale for this approach is that most bookings identified through association have zero children. Customers with special requests also tend to have zero children, and most bookings, according to the data, are for room A (2 adults). Therefore, targeting the "children" segment too heavily would be a waste of resources.

Business recommendation based on hierarchical clustering

In this section I will name each cluster for the hotel to have more focus on the customer clusters and behavior.

Cluster 0: City Families/Couples

- **Recommendation:** Develop "City Family Fun" packages specifically for city hotels.

These packages should cater to the interests of families or couples, offering benefits such as tickets to city attractions or family meal deals. Furthermore, implement flexible pricing strategies. Consider charging slightly higher rates during peak seasons, while offering attractive discounts for extended stays or during off-peak periods to incentivize bookings.

Cluster 1: Early Bird Cancellers

- **Recommendation:** Investigate the reasons behind booking cancellations by surveying customers who have canceled their reservations. This feedback will provide valuable insights into how to reduce cancellations in the future. Additionally, consider offering booking options that allow for easy date modifications. This flexibility can instill greater confidence in customers and encourage them to book with the hotel well in advance.

Cluster 2: Value Resort Seekers

- **Recommendation:** Promote hotels located outside the city center by increasing their visibility. Emphasize their affordability and the relaxing atmosphere they provide. Highlight amenities such as beautiful gardens, swimming pools, and peaceful surroundings in hotel advertisements.

Cluster 3: Parking & Request People

- **Recommendation:** Clearly and prominently advertise the availability of convenient parking at the hotel. Additionally, enhance staff training to ensure exceptional handling of special requests. Recognizing the importance of personalized service to these customers, prioritize making them feel heard, understood, and well-assisted.

Cluster 4: Luxury Family Spenders

- **Recommendation:** Introduce exclusive "Luxury Family Getaways" packages tailored for families staying at family-friendly hotels. These packages should emphasize top-tier service and unique experiences that cater to the preferences of affluent families. Given the potential for unforeseen circumstances, consider implementing flexible cancellation policies for these bookings to accommodate possible plan changes.

Cluster 5: Loyal & Cheap

- **Recommendation:** Enhance loyalty programs to provide even greater rewards for returning customers. Streamline the booking process for loyal guests to ensure a fast and effortless experience. Furthermore, offer attractive rates, particularly for business travelers, to incentivize repeat bookings.

Cluster 6: Solo City Fans

- **Recommendation:** Develop "Solo City Explorer" deals specifically designed for solo travelers seeking urban accommodations. Emphasize the prime location and safety of the hotels in your advertising. Recognizing their reliability and low cancellation rates, reward these dependable customers with exclusive perks.

Overall recommendation for hotel manager:

- **Implement Dynamic Pricing Strategies:** Avoid deep discounts for very early bookings; instead, offer moderate savings for booking slightly in advance. Introduce attractive promotions specifically for families and for stays during the summer season to boost occupancy.

- **Refine Targeted Advertising:** Tailor advertising campaigns to specific demographics. Showcase family-friendly rooms to families, romantic packages to couples, and city hotels to urban enthusiasts. Leverage social media to reach these target audiences effectively.
- **Proactive Customer Communication:** Engage with customers considering cancellation to understand their concerns and encourage them to maintain their bookings. Facilitate easy date modifications. Provide exceptional service to guests requiring parking or with special requests. Reward repeat customers for their continued loyalty.
- **Optimize Advertising Spend:** Avoid inefficient advertising spend on demographics with historically low conversion rates (e.g., large families, if data suggests low booking rates). Prioritize securing bookings with a lower probability of cancellation.
- **Develop Specialized Offers and Ancillary Revenue Streams:** Curate tailored packages for families, solo travelers, and luxury guests. Actively promote meal plans and parking options to increase ancillary revenue for the hotel.

7.2 Next Step

- **Implementing Neural Networks for detecting hotel cancellation:** Neural Networks (NN) offer a promising approach to understanding cancellation patterns with greater depth, especially as a relatively new method. While conventional models like decision trees (currently the best performing) still face uncertainty in identifying potential cancellations, implementing NN could lead to more accurate predictions and improved results.
- **Requiring more dynamic data:** The current dataset is significantly reduced due to a high percentage (26%) of duplicate records. This is concerning, particularly given the

claim in the original paper that these records were directly extracted from an SQL database. Further investigation into the data itself is crucial to identify the source of these duplicates and prevent potential bias in the analysis.

- **Acquiring better setup for SVM implementation:** The Support Vector Machine (SVM) model is a powerful tool for this implementation and has emerged as one of the top performers in my analysis. With increased computational resources, I believe this model has the potential to become the leading classification model, rivaling the performance of Neural Networks.

7.3 Methodological Reflection

Regression

During the Ordinary Least Squares (OLS) model analysis, I encountered both technical and knowledge-based challenges. My limited experience with regression models, specifically in areas like the Durbin-Watson test, Breusch-Pagan (BP) test, or Shapiro-Wilk (SW) test for heteroscedasticity and autocorrelation, posed a significant learning curve. I realized the value of these tests, particularly for time series data where non-normal residuals and autocorrelation are common. The abundance of available time series analysis models can make selecting the most appropriate one difficult. Fortunately, AI tools have enabled me to quickly acquire the necessary knowledge, mitigating my concerns about the coding aspects.

Clustering

I found the clustering tasks less challenging, likely because my OLS analysis deepened my understanding of the data and the meaning of each variable. Current AI technology is invaluable

for simplifying complex and tedious coding tasks, allowing us to focus on implementation.

However, a strong familiarity with the data itself remains essential for effectively interpreting and communicating insights. Understanding the underlying concepts and nuances of the data is something that AI cannot currently replace, at least not entirely.

Association

Association rule mining proved to be quite interesting and effective. However, I observed that increasing the number of input variables led to many seemingly duplicate rules. While technically distinct due to variations in performance metrics when itemset positions are swapped (e.g., "if you buy butter, you will buy milk" vs. "if you buy milk, you will buy butter"), they essentially represent the same association. The lack of causal directionality in association rules means the model generates numerous combinations of items, making it challenging to identify genuinely valuable and interpretable rules. This is an area I believe I can improve upon.

Classification

The classification model analysis is excellent because, by implementing these models, I learned that each model has its own limitations and strengths depending on the dataset. Some models, like decision trees, are very computationally efficient, while others, like SVM, are more powerful but require significantly more computational power. Additionally, some model performances are highly dependent on the chosen variables, as different types of variables will affect the model choice. In real-world datasets (e.g., hotel datasets), there are often mixed types of variables that require further transformation, such as standardization. This is crucial to prevent unreasonable distance calculations, especially for SVM, where binary variables (0 and 1) cannot be easily compared to continuous variables in a range of, say, 1 to 1000. The distance between

these two types of variables can be huge, and that is a key lesson I've learned from this experience.

Appendix

```

--- HAC Robust Model Summary ---
                                OLS Regression Results
=====
Dep. Variable:                  adr      R-squared:                  0.608
Model:                        OLS      Adj. R-squared:            0.608
Method:                      Least Squares      F-statistic:              2451.
Date:                        Tue, 11 Mar 2025      Prob (F-statistic):        0.00
Time:                        20:45:44      Log-Likelihood:            -3.3603e+05
No. Observations:            71368      AIC:                      6.722e+05
Df Residuals:                71306      BIC:                      6.727e+05
Df Model:                    61
Covariance Type:            HAC

```

	coef	std err	t	P> t	[0.025	0.975]
const	67.5790	3.161	21.378	0.000	61.383	73.775
is_canceled	5.8480	0.807	7.248	0.000	4.267	7.429
lead_time	-0.1219	0.005	-26.304	0.000	-0.131	-0.113
arrival_date_day_of_month	0.1246	0.038	3.284	0.001	0.050	0.199
stays_in_weekend_nights	0.3621	0.295	1.227	0.220	-0.216	0.940
stays_in_week_nights	1.4228	0.161	8.857	0.000	1.108	1.738
adults	13.1670	0.378	34.846	0.000	12.426	13.908
children	12.9140	0.413	31.294	0.000	12.105	13.723
is_repeated_guest	-12.9108	0.990	-13.039	0.000	-14.852	-10.970
previous_cancellations	0.9620	0.632	1.523	0.128	-0.276	2.200
previous_bookings_not_canceled	-0.3593	0.092	-3.906	0.000	-0.540	-0.179
booking_changes	0.8727	0.204	4.288	0.000	0.474	1.272
days_in_waiting_list	0.2055	0.047	4.337	0.000	0.113	0.298
required_car_parking_spaces	6.7787	0.584	11.598	0.000	5.633	7.924
total_of_special_requests	1.2319	0.190	6.498	0.000	0.860	1.604
is_city_hotel	28.0715	1.025	27.374	0.000	26.062	30.081
meal_FB	41.8849	2.525	16.590	0.000	36.936	46.833
meal_HB	24.4189	0.649	37.618	0.000	23.147	25.691
meal_SC	-13.5760	0.395	-34.332	0.000	-14.351	-12.801
meal_undefined	35.1586	3.065	11.471	0.000	29.151	41.166
market_segment_Aviation	-10.6083	1.856	-5.717	0.000	-14.245	-6.971
market_segment_Complementary	-101.5684	1.756	-57.831	0.000	-105.011	-98.126
market_segment_Corporate	-16.6735	0.832	-20.041	0.000	-18.304	-15.043
market_segment_Direct	-2.4960	0.492	-5.071	0.000	-3.461	-1.531
market_segment_Groups	-26.1489	1.857	-14.082	0.000	-29.789	-22.509
market_segment_Offline TA/TO	-24.6649	0.523	-47.201	0.000	-25.689	-23.641

market_segment_Undefined	-121.0135	4.793	-25.247	0.000	-130.408	-111.619
assigned_room_type_B	-4.1094	0.959	-4.286	0.000	-5.989	-2.230
assigned_room_type_C	6.9331	1.001	6.926	0.000	4.971	8.895
assigned_room_type_D	9.0902	0.335	27.125	0.000	8.433	9.747
assigned_room_type_E	18.4425	0.560	32.961	0.000	17.346	19.539
assigned_room_type_F	26.5916	0.843	31.543	0.000	24.939	28.244
assigned_room_type_G	26.7904	1.227	21.841	0.000	24.386	29.195
assigned_room_type_H	24.6218	1.982	12.422	0.000	20.737	28.507
assigned_room_type_I	-37.1734	3.370	-11.029	0.000	-43.779	-30.567
assigned_room_type_K	-32.4888	3.867	-8.402	0.000	-40.068	-24.910
assigned_room_type_L	-89.8486	2.680	-33.524	0.000	-95.102	-84.596
assigned_room_type_P	-11.5104	24.415	-0.471	0.637	-59.364	36.344
deposit_type_No Deposit	-11.7774	2.047	-5.754	0.000	-15.789	-7.765
deposit_type_Refundable	10.3372	8.777	1.178	0.239	-6.866	27.541
customer_type_Contract	-1.7117	1.310	-1.307	0.191	-4.279	0.856
customer_type_Group	-1.5675	1.545	-1.015	0.310	-4.595	1.460
customer_type_Transient	-1.7033	0.660	-2.581	0.010	-2.997	-0.410
year_2015	-25.8447	1.910	-13.529	0.000	-29.589	-22.100
year_2016	-12.8856	0.798	-16.141	0.000	-14.450	-11.321
month_1	-20.2740	1.732	-11.702	0.000	-23.670	-16.878
month_2	-17.0547	1.645	-10.369	0.000	-20.278	-13.831
month_3	-8.5673	1.630	-5.257	0.000	-11.761	-5.373
month_4	10.0932	1.629	6.198	0.000	6.901	13.285
month_5	22.2244	1.725	12.882	0.000	18.843	25.606
month_6	29.3926	1.653	17.785	0.000	26.153	32.632
month_7	37.7495	2.223	16.978	0.000	33.392	42.107
month_8	41.4606	2.527	16.404	0.000	36.507	46.414
month_9	36.9071	1.895	19.478	0.000	33.193	40.621
month_10	15.4161	1.702	9.060	0.000	12.081	18.751
month_10	15.4161	1.702	9.060	0.000	12.081	18.751
month_11	-2.2527	1.949	-1.156	0.248	-6.072	1.566
week_0	-1.3556	0.761	-1.782	0.075	-2.847	0.136
week_1	-1.9296	0.864	-2.233	0.026	-3.623	-0.236
week_2	-2.0833	0.871	-2.392	0.017	-3.790	-0.376
week_3	-1.8721	0.903	-2.073	0.038	-3.642	-0.102
week_4	-1.6243	0.862	-1.885	0.059	-3.314	0.065
week_5	-1.8674	0.831	-2.246	0.025	-3.497	-0.238
=====						
Omnibus:	4307.620	Durbin-Watson:		1.161		
Prob(Omnibus):	0.000	Jarque-Bera (JB):		19381.128		
Skew:	0.046	Prob(JB):		0.00		
Kurtosis:	5.551	Cond. No.		2.16e+04		
=====						

Figure 11.1: OLSHAC model performance